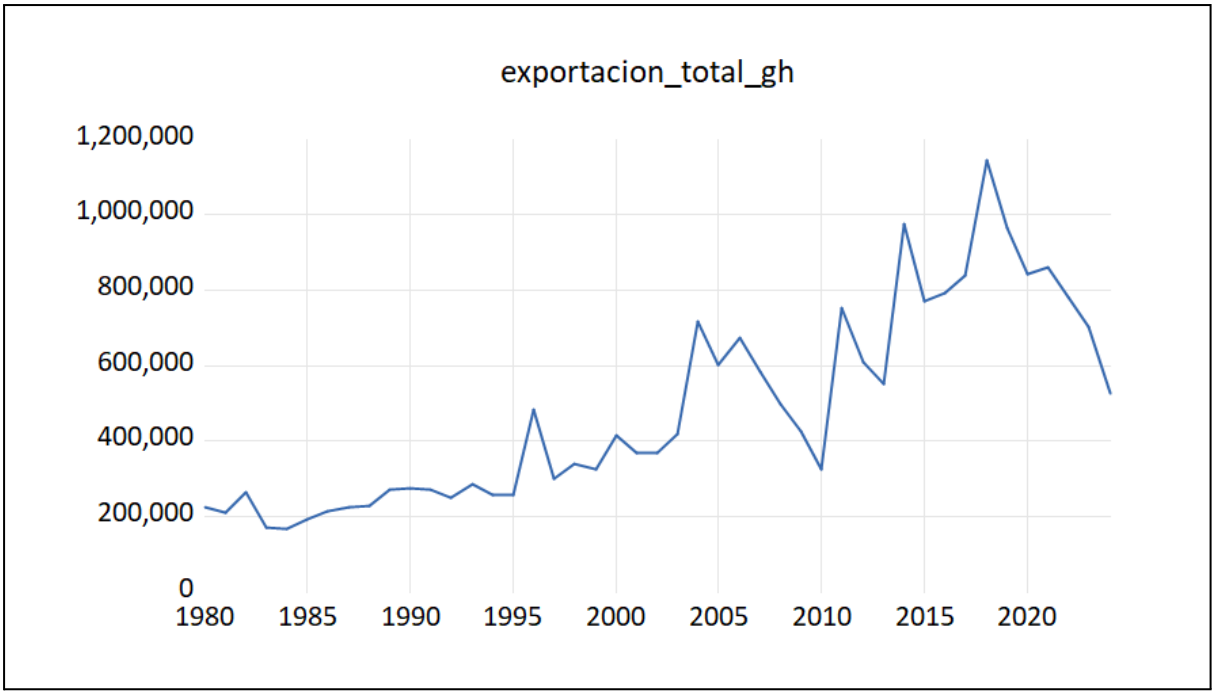


El gráfico crece muy lentamente entre 1980-2000, luego se ve una subida con una gran volatilidad, con dos grandes caídas entre 2006-2010 y 2018-2024.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: EXPORTACION TOTAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.756365	0.0287
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

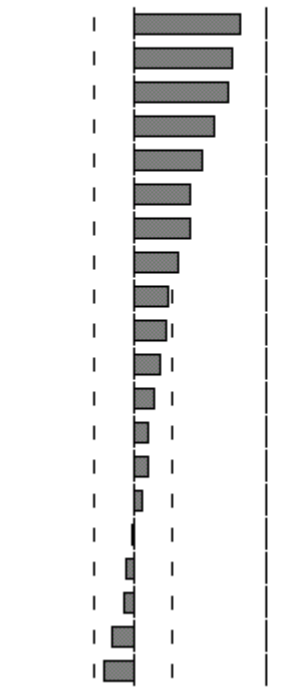
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXPORTACION TOTAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:31
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPORTACION TOTAL GH(-1)	-0.582623	0.155103	-3.756365	0.0005
C	64884.70	40058.45	1.619751	0.1130
@TREND("1980")	9894.994	3182.913	3.108785	0.0034

R-squared	0.258826	Mean dependent var	6913.498
Adjusted R-squared	0.222671	S.D. dependent var	142028.8
S.E. of regression	125221.5	Akaike info criterion	26.37930
Sum squared resid	6.43E+11	Schwarz criterion	26.50095
Log likelihood	-577.3446	Hannan-Quinn criter.	26.42441
F-statistic	7.158810	Durbin-Watson stat	1.952845
Prob(F-statistic)	0.002155		

Como la tendencia determinística resulta significativo analizo el p-valor para verificar la presencia de raíz unitaria al 5% de significatividad, como el p-valor es 0.0287<0.05, Rechazo la hipótesis nula de raíz unitaria y por lo tanto analizo la serie en niveles.

Date: 10/14/26 Time: 18:32
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.843	0.843	34.164	0.000
		2	0.778	0.232	63.913	0.000
		3	0.739	0.147	91.433	0.000
		4	0.646	-0.139	112.99	0.000
		5	0.540	-0.183	128.43	0.000
		6	0.455	-0.088	139.67	0.000
		7	0.446	0.261	150.74	0.000
		8	0.360	-0.083	158.16	0.000
		9	0.277	-0.124	162.65	0.000
		10	0.258	0.029	166.69	0.000
		11	0.209	-0.026	169.41	0.000
		12	0.164	0.053	171.13	0.000
		13	0.120	-0.017	172.08	0.000
		14	0.110	-0.007	172.91	0.000
		15	0.069	-0.092	173.25	0.000
		16	-0.006	-0.138	173.25	0.000
		17	-0.053	-0.117	173.46	0.000
		18	-0.081	0.074	173.98	0.000
		19	-0.166	-0.148	176.23	0.000
		20	-0.223	-0.035	180.45	0.000

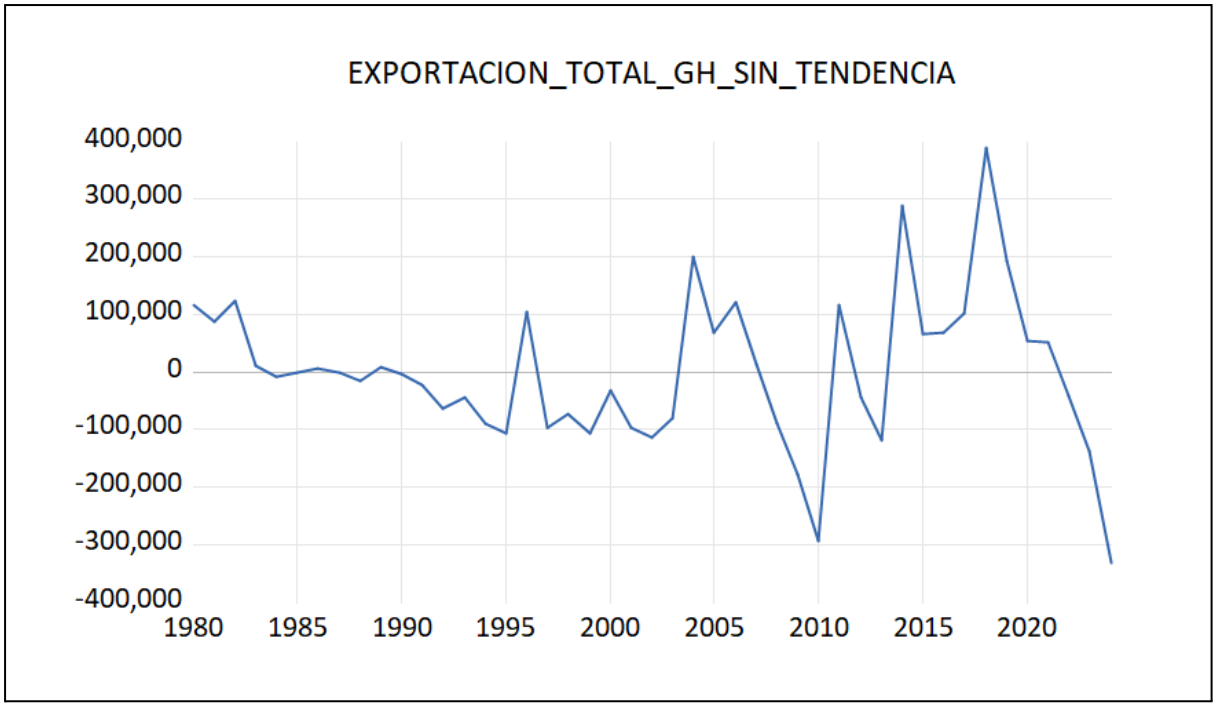
Compruebo la significatividad de la tendencia:

Dependent Variable: EXPORTACION TOTAL GH
Method: Least Squares
Date: 10/14/26 Time: 18:37
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	106537.9	39278.73	2.712355	0.0096
@TREND	17096.48	1537.488	11.11975	0.0000

R-squared	0.741972	Mean dependent var	482660.4
Adjusted R-squared	0.735972	S.D. dependent var	260679.8
S.E. of regression	133946.9	Akaike info criterion	26.49170
Sum squared resid	7.71E+11	Schwarz criterion	26.57200
Log likelihood	-594.0633	Hannan-Quinn criter.	26.52163
F-statistic	123.6488	Durbin-Watson stat	1.130227
Prob(F-statistic)	0.000000		

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:



Date: 10/14/26 Time: 18:39
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.355	0.355	6.0601	0.014
		2	0.168	0.049	7.4566	0.024
		3	0.110	0.041	8.0679	0.045
		4	-0.058	-0.133	8.2419	0.083
		5	-0.267	-0.256	12.024	0.034
		6	-0.329	-0.195	17.903	0.006
		7	0.002	0.278	17.903	0.012
		8	-0.100	-0.094	18.477	0.018
		9	-0.205	-0.224	20.935	0.013
		10	-0.072	-0.082	21.246	0.019
		11	0.059	0.090	21.464	0.029
		12	0.011	0.048	21.473	0.044
		13	0.018	0.061	21.495	0.064
		14	0.227	0.062	25.022	0.034
		15	0.125	-0.134	26.123	0.037
		16	-0.050	-0.101	26.307	0.050
		17	-0.187	-0.191	28.938	0.035
		18	-0.104	0.015	29.782	0.040
		19	-0.207	-0.053	33.267	0.022
		20	-0.213	0.006	37.100	0.011

Se ve como cambia el correlograma con respecto al anterior, tanto la FAC como la FACP dejan de ser significativas en el segundo rezago

Dependent Variable: EXPORTACION_TOTAL_GH_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/16/26 Time: 22:20
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Failure to improve likelihood (non-zero gradients) after 17 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.562156	0.308919	1.819752	0.0759
MA(1)	-0.181383	0.361233	-0.502121	0.6182

R-squared	0.154563	Mean dependent var	-2665.806
Adjusted R-squared	0.134433	S.D. dependent var	132719.8
S.E. of regression	123476.9	Akaike info criterion	26.32989
Sum squared resid	6.40E+11	Schwarz criterion	26.41099
Log likelihood	-577.2575	Hannan-Quinn criter.	26.35996
Durbin-Watson stat	1.880576		

Inverted AR Roots	.56
Inverted MA Roots	.18

El componente MA(1) no resultó significativo, por lo que pruebo la estimación solo con el componente AR(1)

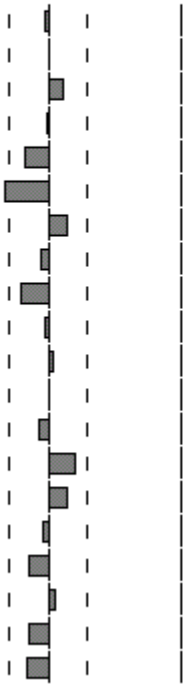
Dependent Variable: EXPORTACION_TOTAL_GH_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 18:44
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 10 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.413738	0.150447	2.750058	0.0087

R-squared	0.149222	Mean dependent var	-2665.806
Adjusted R-squared	0.149222	S.D. dependent var	132719.8
S.E. of regression	122417.6	Akaike info criterion	26.29073
Sum squared resid	6.44E+11	Schwarz criterion	26.33128
Log likelihood	-577.3960	Hannan-Quinn criter.	26.30577
Durbin-Watson stat	1.941199		

Inverted AR Roots	.41

Date: 10/14/26 Time: 18:44
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.030	-0.030	0.0427	
		2	0.015	0.015	0.0542	0.816
		3	0.122	0.123	0.7864	0.675
		4	-0.007	0.001	0.7886	0.852
		5	-0.178	-0.185	2.4262	0.658
		6	-0.338	-0.384	8.5242	0.130
		7	0.140	0.128	9.6028	0.142
		8	-0.057	0.047	9.7823	0.201
		9	-0.218	-0.162	12.523	0.129
		10	-0.019	-0.155	12.545	0.184
		11	0.039	-0.092	12.640	0.245
		12	0.003	-0.005	12.641	0.317
		13	-0.073	0.038	12.992	0.370
		14	0.205	0.127	15.833	0.258
		15	0.144	0.011	17.282	0.241
		16	-0.040	-0.076	17.396	0.296
		17	-0.141	-0.271	18.892	0.274
		18	0.063	-0.019	19.200	0.317
		19	-0.140	-0.083	20.795	0.290
		20	-0.164	0.007	23.062	0.235

La estimación del coeficiente del AR(1) resulto significativa, con ruido blanco, y raíz dentro del círculo unitario, sin embargo debo probar la estimación solo con el componente MA(1)

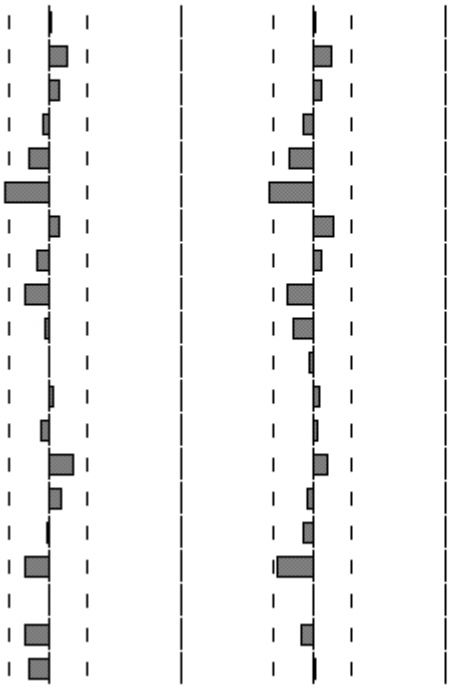
Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 18:42
Sample: 1980 2024
Included observations: 45
Failure to improve likelihood (non-zero gradients) after 14 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1979

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.365908	0.146871	2.491356	0.0166

R-squared	0.126457	Mean dependent var	1.24E-10
Adjusted R-squared	0.126457	S.D. dependent var	132416.0
S.E. of regression	123760.7	Akaike info criterion	26.31206
Sum squared resid	6.74E+11	Schwarz criterion	26.35221
Log likelihood	-591.0213	Hannan-Quinn criter.	26.32703
Durbin-Watson stat	1.798459		

Inverted MA Roots	-.37

Date: 10/14/26 Time: 18:43
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.032	0.032	0.0479	
		2	0.151	0.150	1.1684	0.280
		3	0.082	0.075	1.5090	0.470
		4	-0.042	-0.070	1.5987	0.660
		5	-0.155	-0.183	2.8715	0.580
		6	-0.336	-0.343	8.9995	0.109
		7	0.083	0.161	9.3828	0.153
		8	-0.085	0.072	9.7981	0.200
		9	-0.181	-0.197	11.731	0.164
		10	-0.030	-0.141	11.785	0.226
		11	0.015	-0.030	11.799	0.299
		12	0.038	0.056	11.890	0.372
		13	-0.052	0.042	12.071	0.440
		14	0.195	0.107	14.666	0.329
		15	0.106	-0.048	15.453	0.348
		16	-0.006	-0.079	15.456	0.419
		17	-0.177	-0.272	17.810	0.335
		18	0.010	0.002	17.817	0.400
		19	-0.182	-0.085	20.509	0.305
		20	-0.141	0.021	22.198	0.275

Como con ambos modelos llego a coeficientes significativos, ruido blanco y raíz dentro del circulo unitario, utilizo el criterio de informacion de Akaike para comparar entre modelos. Debido a que al estimar el AR(1) se resta una observacion, y el AIC requiere la misma cantidad de observaciones debo estimar el modelo MA(1) nuevamente con 44 obs.

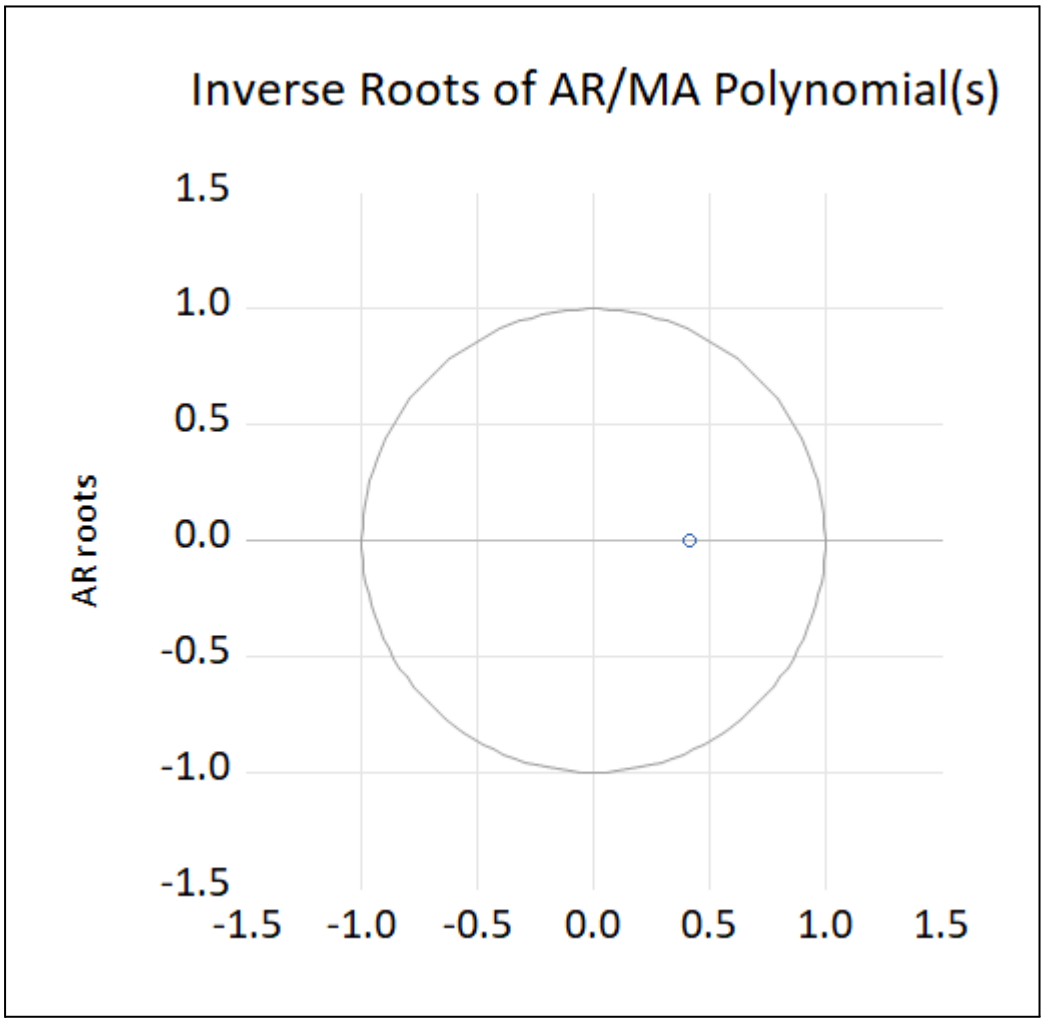
Dependent Variable: EXPORTACION TOTAL GH SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 18:45
Sample: 1981 2024
Included observations: 44
Failure to improve likelihood (non-zero gradients) after 15 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

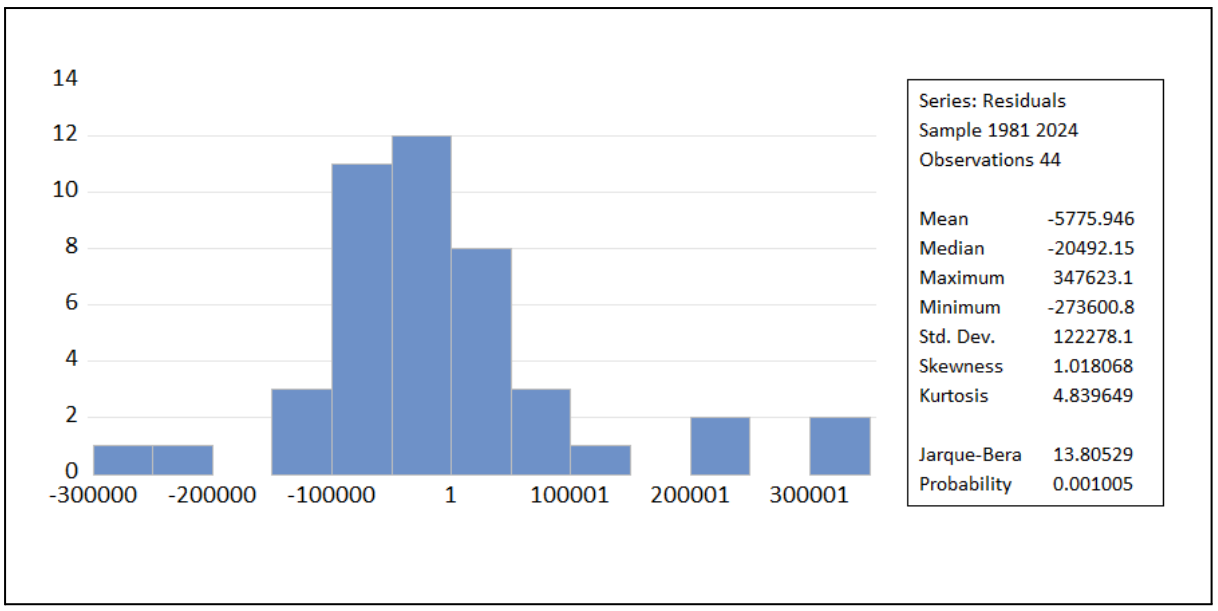
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.360762	0.149673	2.410333	0.0203

R-squared	0.122334	Mean dependent var	-2665.806
Adjusted R-squared	0.122334	S.D. dependent var	132719.8
S.E. of regression	124337.0	Akaike info criterion	26.32184
Sum squared resid	6.65E+11	Schwarz criterion	26.36239
Log likelihood	-578.0806	Hannan-Quinn criter.	26.33688
Durbin-Watson stat	1.796947		

Inverted MA Roots	-.36

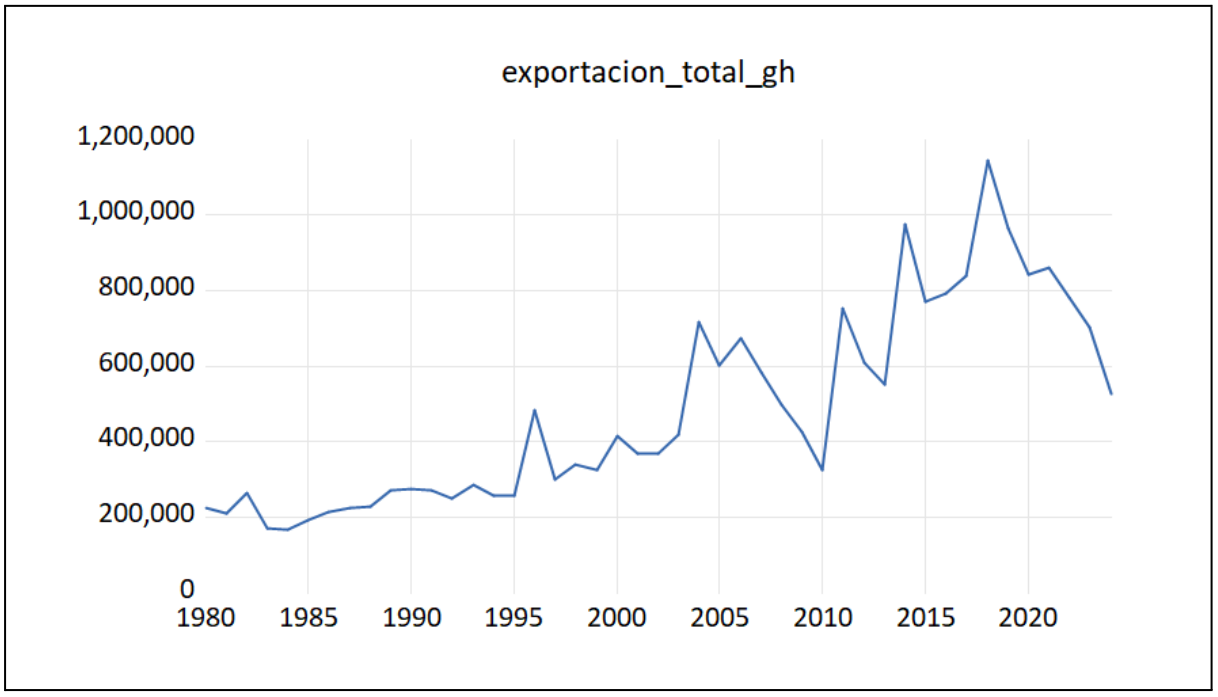
Al comparar el AIC vemos que el modelo AR(1) AIC=26.29, y en el modelo MA(1) AIC=26.32, por lo tanto elijo el modelo con el criterio mas bajo, es decir, el AR(1)





El test de Jarque-Bera del modelo AR(1) rechaza la hipótesis nula de que los residuos se comportan normalmente.

Incorporando una variable de impacto:



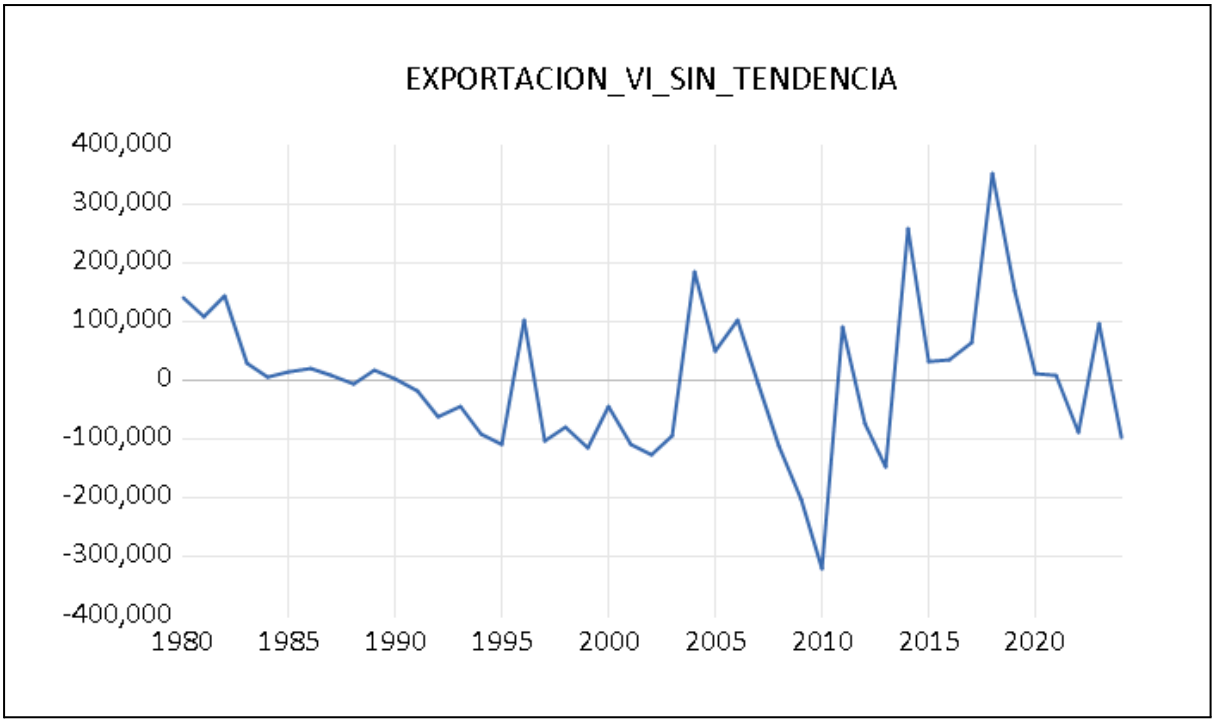
Durante los últimos años está ocurriendo una crisis respecto a la producción del cacao en África Occidental, provocada por un virus sobre la planta de cacao, que afectó a Ghana y Costa de Marfil, dos de los mayores productores de cacao a nivel mundial (60% de la oferta mundial).
Es por eso que decidí incorporar una variable de intervencion que refleje ese shock tanto en la producción como en el precio internacional, y por lo tanto en el precio del productor.
Con respecto a la variable exportacion total

Dependent Variable: EXPORTACION_TOTAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:34
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI_CRISIS	-281198.5	95567.65	-2.942403	0.0053
C	83987.63	36990.97	2.270490	0.0284
@TREND	18689.57	1516.467	12.32441	0.0000

R-squared	0.786071	Mean dependent var	482660.4
Adjusted R-squared	0.775884	S.D. dependent var	260679.8
S.E. of regression	123408.2	Akaike info criterion	26.34872
Sum squared resid	6.40E+11	Schwarz criterion	26.46917
Log likelihood	-589.8463	Hannan-Quinn criter.	26.39362
F-statistic	77.16340	Durbin-Watson stat	1.403986
Prob(F-statistic)	0.000000		

La variable de intervención que mejor refleja la crisis en África occidental es aquella que incluye a los anios 2023 y 2024.



Null Hypothesis: EXPORTACION_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

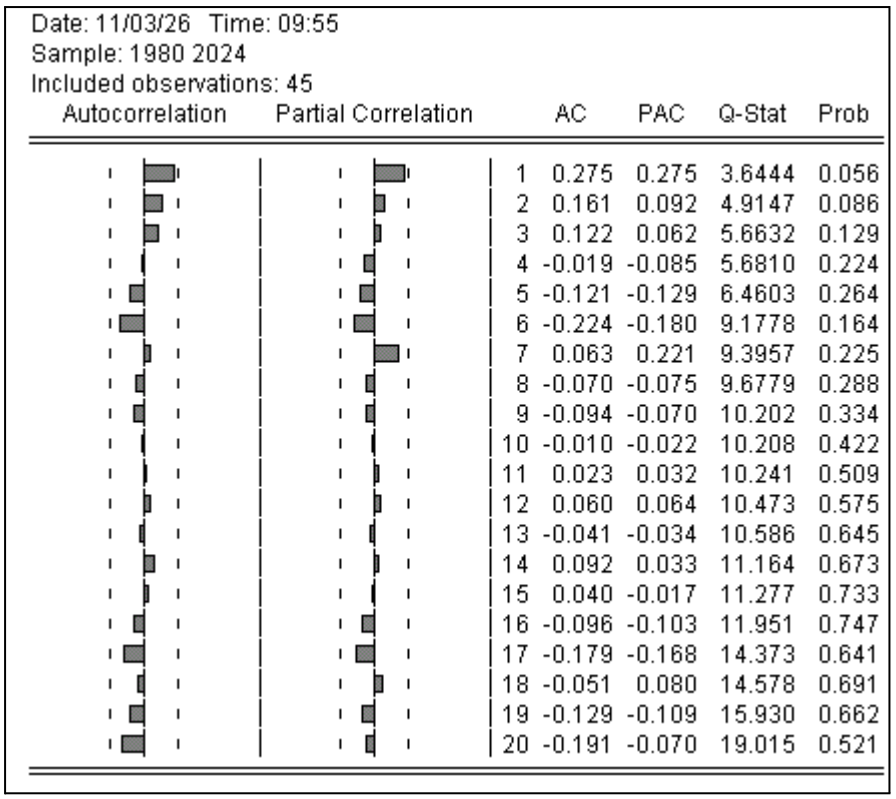
		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.964373	0.0000
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

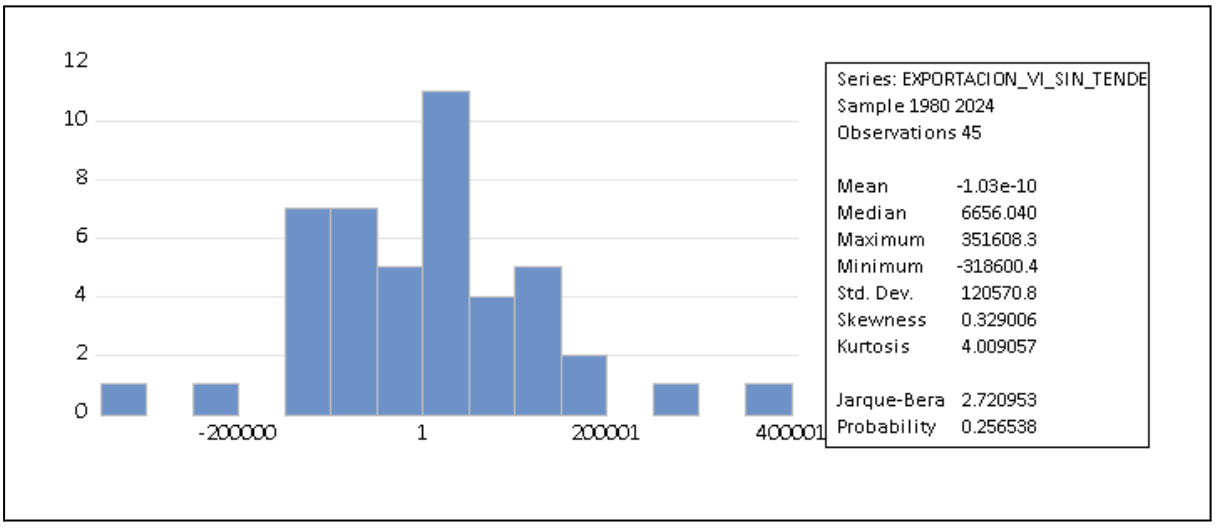
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXPORTACION_VI_SIN_TENDENCIA)
Method: Least Squares
Date: 11/02/26 Time: 12:36
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXPORTACION_VI_SIN_TENDENCIA(-1)	-0.720531	0.145140	-4.964373	0.0000

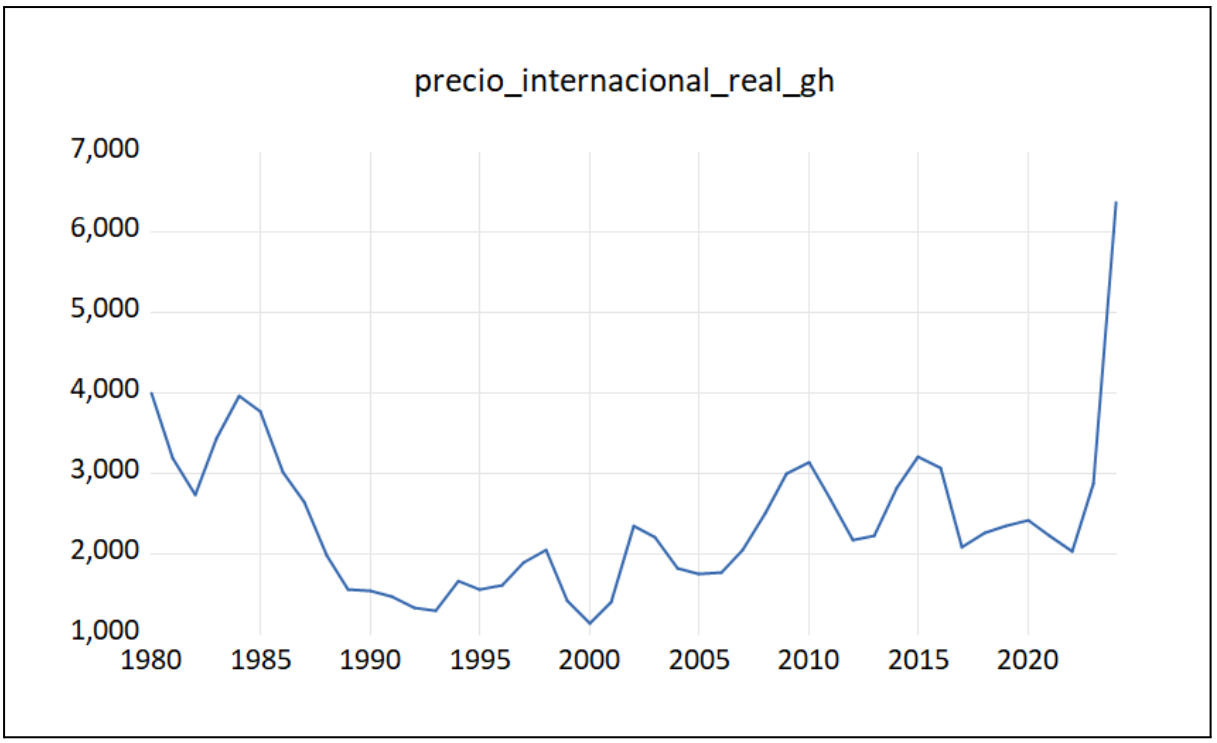
R-squared	0.363424	Mean dependent var	-5385.196
Adjusted R-squared	0.363424	S.D. dependent var	144413.2
S.E. of regression	115221.1	Akaike info criterion	26.16956
Sum squared resid	5.71E+11	Schwarz criterion	26.21011
Log likelihood	-574.7303	Hannan-Quinn criter.	26.18460
Durbin-Watson stat	2.048261		



	AC	PAC	Q-Stat	Prob
1	0.275	0.275	3.6444	0.056
2	0.161	0.092	4.9147	0.086
3	0.122	0.062	5.6632	0.129
4	-0.019	-0.085	5.6810	0.224
5	-0.121	-0.129	6.4603	0.264
6	-0.224	-0.180	9.1778	0.164
7	0.063	0.221	9.3957	0.225
8	-0.070	-0.075	9.6779	0.288
9	-0.094	-0.070	10.202	0.334
10	-0.010	-0.022	10.208	0.422
11	0.023	0.032	10.241	0.509
12	0.060	0.064	10.473	0.575
13	-0.041	-0.034	10.586	0.645
14	0.092	0.033	11.164	0.673
15	0.040	-0.017	11.277	0.733
16	-0.096	-0.103	11.951	0.747
17	-0.179	-0.168	14.373	0.641
18	-0.051	0.080	14.578	0.691
19	-0.129	-0.109	15.930	0.662
20	-0.191	-0.070	19.015	0.521



Al ver el correlograma luego de eliminar la tendencia, el intercepto y el efecto de la crisis de los anios 23/24 arroja directamente ruido blanco, por lo que no hay ningun componente AR/MA que incluir en el modelo.
Además el comportamiento de la nueva serie no rechaza el test de normalidad.



En el gráfico se observa una fuerte caída entre 1985 y 1990, luego cierta estabilidad, y finalmente una gran subida en los últimos años del precio internacional del cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.243925	0.8882
Test critical values:	1% level	-4.186481	
	5% level	-3.518090	
	10% level	-3.189732	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.172549	0.138713	-1.243925	0.2210
D(PRECIO INTERNACIONAL REAL GH	0.616396	0.226211	2.724873	0.0096
C	221.7287	385.0076	0.575907	0.5680
@TREND("1980")	11.30600	7.774911	1.454164	0.1539

R-squared	0.240057	Mean dependent var	74.03259
Adjusted R-squared	0.181600	S.D. dependent var	682.1569
S.E. of regression	617.1163	Akaike info criterion	15.77640
Sum squared resid	14852469	Schwarz criterion	15.94023
Log likelihood	-335.1926	Hannan-Quinn criter.	15.83682
F-statistic	4.106556	Durbin-Watson stat	1.119953
Prob(F-statistic)	0.012631		

Como la tendencia determinística no resulta significativa (p-valor = 0.1539), vuelvo a hacer el test con intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.566130	0.8673
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.081703	0.144319	-0.566130	0.5746
D(PRECIO INTERNACIONAL REAL GH	0.834834	0.227768	3.665281	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.628098	0.235939	-2.662122	0.0113
C	247.0291	340.6667	0.725134	0.4728

R-squared	0.316966	Mean dependent var	86.39314
Adjusted R-squared	0.263043	S.D. dependent var	685.5347
S.E. of regression	588.5057	Akaike info criterion	15.68344
Sum squared resid	13160881	Schwarz criterion	15.84894
Log likelihood	-325.3523	Hannan-Quinn criter.	15.74410
F-statistic	5.878048	Durbin-Watson stat	1.412468
Prob(F-statistic)	0.002123		

Nuevamente, el intercepto no resulta significativo (p-valor = 0.4728), por lo que realizo el test sin tendencia ni intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: None
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		0.496799	0.8186
Test critical values:	1% level	-2.621185	
	5% level	-1.948886	
	10% level	-1.611932	

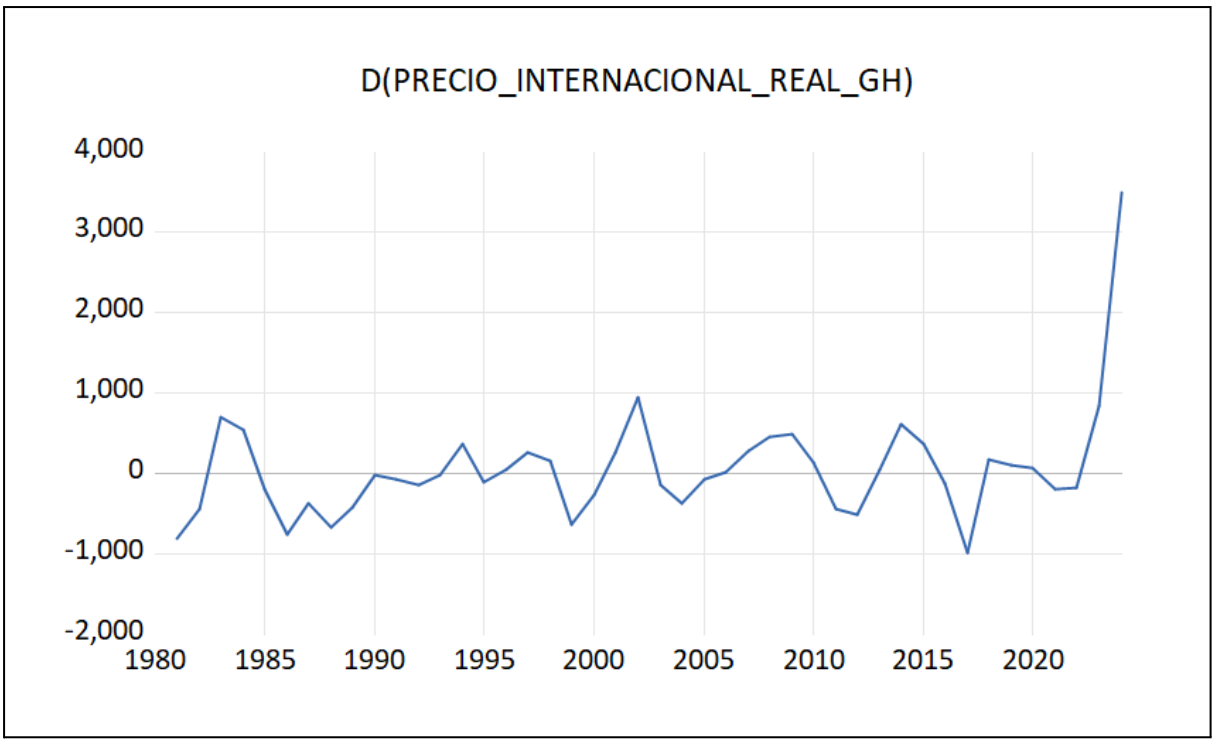
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:50
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	0.019113	0.038472	0.496799	0.6221
D(PRECIO INTERNACIONAL REAL GH	0.802145	0.221901	3.614878	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.682925	0.222133	-3.074394	0.0038

R-squared	0.307515	Mean dependent var	86.39314
Adjusted R-squared	0.272003	S.D. dependent var	685.5347
S.E. of regression	584.9171	Akaike info criterion	15.64957
Sum squared resid	13342993	Schwarz criterion	15.77369
Log likelihood	-325.6409	Hannan-Quinn criter.	15.69506
Durbin-Watson stat	1.441096		

Como el p-valor asociado al test estadístico es 0.8186 no rechazo la hipótesis nula de raíz unitaria por lo que debo diferenciar la serie:



Notar la presencia de un valor atípico en los últimos años (2023-2024) donde el precio internacional se dispara.

Realizo el test de Dickey Fuller nuevamente para asegurarme de que no haya que diferenciar nuevamente la serie:

Null Hypothesis: D(PRECIO_INTERNACIONAL_REAL_GH) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.690253	0.0341
Test critical values:	1% level	-4.192337	
	5% level	-3.520787	
	10% level	-3.191277	

*MacKinnon (1996) one-sided p-values.

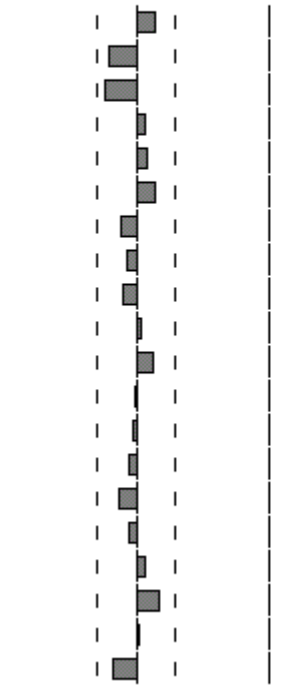
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIONAL_REAL_GH,2)
Method: Least Squares
Date: 10/20/26 Time: 17:58
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(PRECIO_INTERNACIONAL_REAL_GH(-1	-0.959795	0.260089	-3.690253	0.0007
D(PRECIO_INTERNACIONAL_REAL_GH(-1	0.716317	0.215317	3.326810	0.0020
C	-293.3396	192.6193	-1.522898	0.1361
@TREND("1980")	14.97777	7.259097	2.063311	0.0460

R-squared	0.314819	Mean dependent var	93.76005
Adjusted R-squared	0.260726	S.D. dependent var	651.7977
S.E. of regression	560.4226	Akaike info criterion	15.58565
Sum squared resid	11934794	Schwarz criterion	15.75114
Log likelihood	-323.2987	Hannan-Quinn criter.	15.64631
F-statistic	5.819927	Durbin-Watson stat	1.563827
Prob(F-statistic)	0.002247		

Como la tendencia resulta significativa y el estadístico del test tiene un p-valor = 0.0341, rechazo la hipótesis nula de raíz unitaria en la serie diferenciada, al 5% de significatividad.

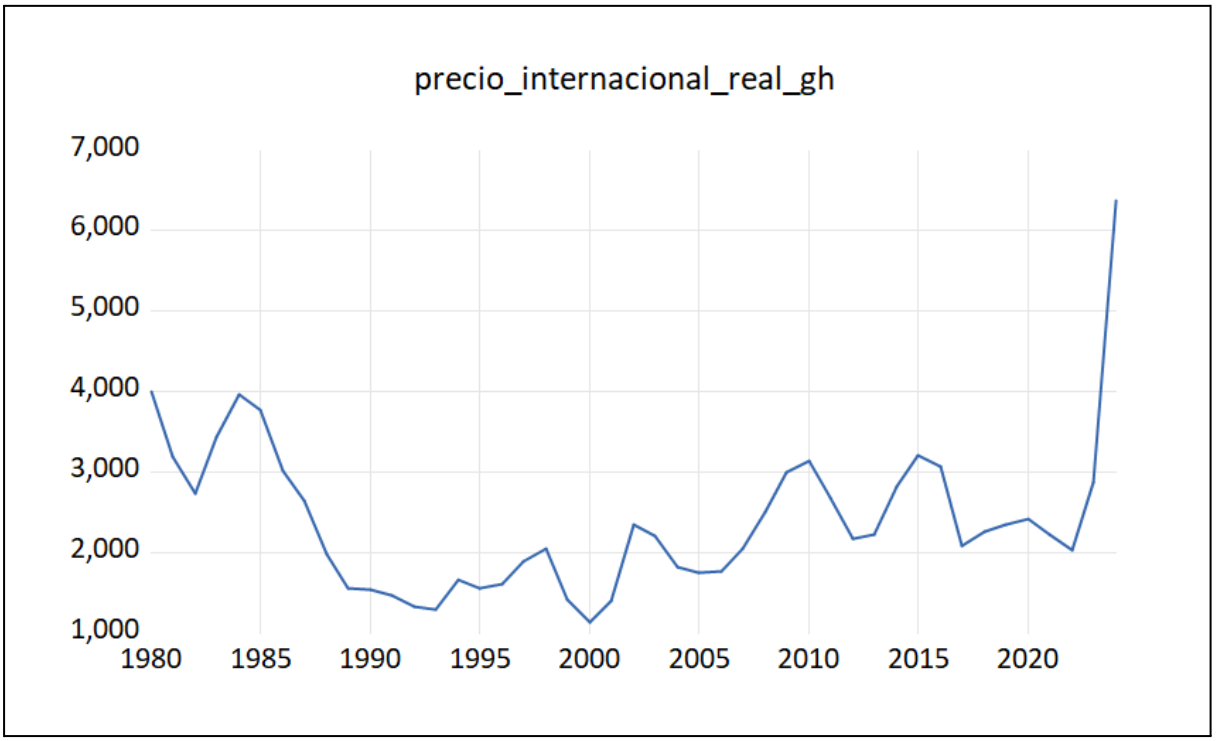
Date: 10/14/26 Time: 18:52
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.145	0.145	0.9644	0.326
		2	-0.219	-0.245	3.2214	0.200
		3	-0.245	-0.185	6.1186	0.106
		4	0.072	0.096	6.3733	0.173
		5	0.084	-0.035	6.7320	0.241
		6	0.148	0.144	7.8796	0.247
		7	-0.110	-0.122	8.5321	0.288
		8	-0.079	0.013	8.8812	0.352
		9	-0.099	-0.091	9.4395	0.398
		10	0.037	-0.017	9.5193	0.484
		11	0.133	0.120	10.582	0.479
		12	-0.000	-0.103	10.582	0.565
		13	-0.017	0.120	10.601	0.644
		14	-0.056	-0.072	10.807	0.701
		15	-0.137	-0.147	12.110	0.671
		16	-0.054	-0.016	12.319	0.722
		17	0.082	-0.035	12.816	0.748
		18	0.181	0.197	15.354	0.638
		19	0.019	-0.059	15.385	0.698
		20	-0.186	-0.075	18.284	0.569

El correlograma de la serie diferenciada se comporta como ruido blanco, lo cual tiene sentido teniendo en cuenta que se trata de la serie de precio internacional del cacao, que al ser una commodity es normal que se comporte como un *random walk*.

Incorporando una variable de intervención:

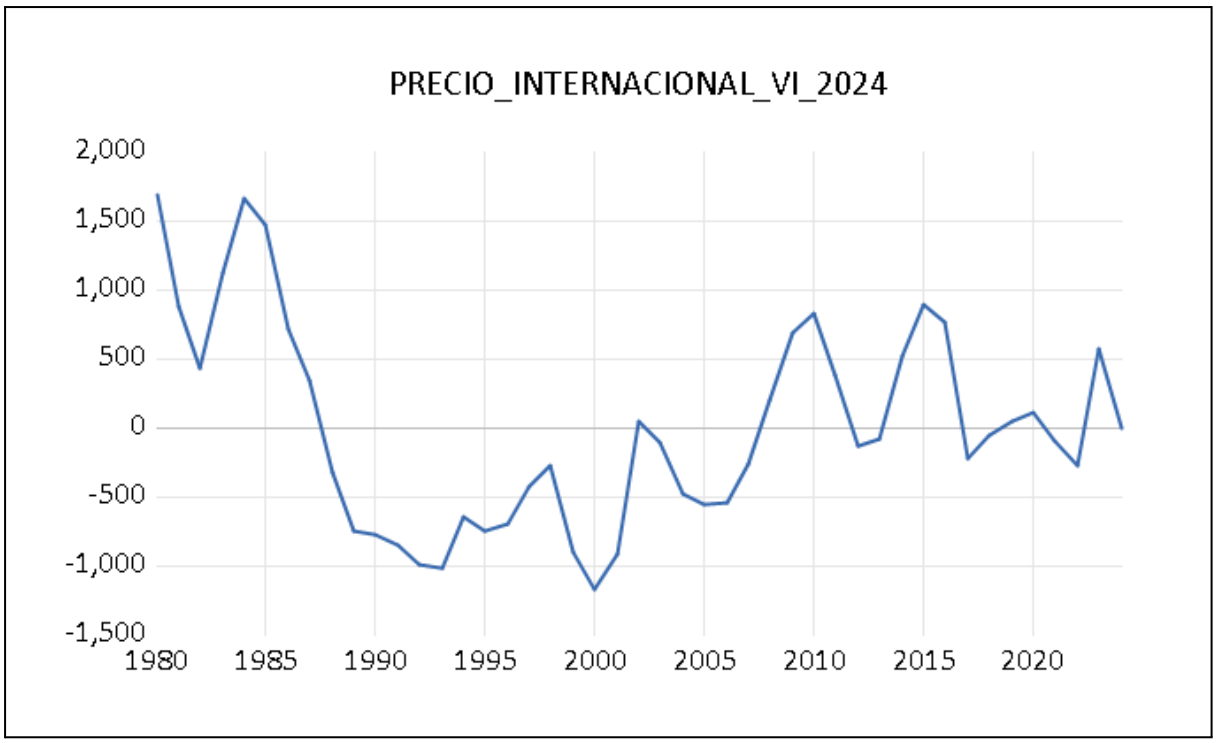


A pesar de que el modelo anterior reportaba un random walk, lo que tiene sentido económico teniendo en cuenta de que se trata de una commoditie, incorporo una variable de intervención debido al pico en el precio internacional en el año 2024.

Dependent Variable: PRECIO_INTERNACIONAL_REAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:49
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2309.389	111.9929	20.62085	0.0000
VI_2024	4058.679	751.2712	5.402415	0.0000

R-squared	0.404317	Mean dependent var	2399.582
Adjusted R-squared	0.390464	S.D. dependent var	951.5185
S.E. of regression	742.8769	Akaike info criterion	16.10236
Sum squared resid	23730240	Schwarz criterion	16.18266
Log likelihood	-360.3032	Hannan-Quinn criter.	16.13230
F-statistic	29.18608	Durbin-Watson stat	0.360362
Prob(F-statistic)	0.000003		



Null Hypothesis: PRECIO_INTERNACIOAL_VI_2024 has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.857434	0.0053
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

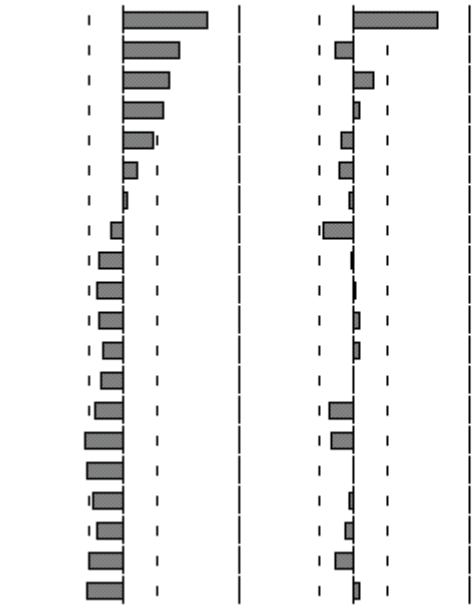
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIOAL_VI_2024)
Method: Least Squares
Date: 11/02/26 Time: 13:00
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO_INTERNACIOAL_VI_2024(-1)	-0.239806	0.083923	-2.857434	0.0066

R-squared	0.153212	Mean dependent var	-38.23194
Adjusted R-squared	0.153212	S.D. dependent var	444.2701
S.E. of regression	408.8221	Akaike info criterion	14.88690
Sum squared resid	7186828.	Schwarz criterion	14.92745
Log likelihood	-326.5119	Hannan-Quinn criter.	14.90194
Durbin-Watson stat	1.412971		

Ahora el test de Dickey Fuller rechaza la hipótesis nula de raíz unitaria, al contrario que el modelo anterior sin variable de intervención.

Date: 11/02/26 Time: 13:01
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.760	0.760	27.778	0.000
		2	0.508	-0.165	40.483	0.000
		3	0.402	0.188	48.620	0.000
		4	0.362	0.047	55.361	0.000
		5	0.270	-0.104	59.204	0.000
		6	0.129	-0.119	60.106	0.000
		7	0.028	-0.030	60.149	0.000
		8	-0.113	-0.270	60.872	0.000
		9	-0.215	-0.018	63.592	0.000
		10	-0.240	0.014	67.076	0.000
		11	-0.211	0.053	69.833	0.000
		12	-0.183	0.049	71.970	0.000
		13	-0.189	-0.004	74.343	0.000
		14	-0.254	-0.216	78.742	0.000
		15	-0.344	-0.197	87.068	0.000
		16	-0.336	-0.000	95.286	0.000
		17	-0.265	-0.026	100.59	0.000
		18	-0.240	-0.069	105.09	0.000
		19	-0.314	-0.161	113.12	0.000
		20	-0.332	0.056	122.42	0.000

Dependent Variable: PRECIO_INTERNACIOAL_VI_2024
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:02
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.760194	0.083923	9.058187	0.0000

R-squared	0.655078	Mean dependent var	-38.23194
Adjusted R-squared	0.655078	S.D. dependent var	696.1040
S.E. of regression	408.8221	Akaike info criterion	14.88690
Sum squared resid	7186828.	Schwarz criterion	14.92745
Log likelihood	-326.5119	Hannan-Quinn criter.	14.90194
Durbin-Watson stat	1.412971		

Inverted AR Roots	.76

Date: 11/02/26 Time: 13:02
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.267	0.267	3.3551	
		2	-0.224	-0.318	5.7717	0.016
		3	-0.290	-0.151	9.9254	0.007
		4	-0.006	0.078	9.9269	0.019
		5	0.110	-0.023	10.561	0.032
		6	0.006	-0.071	10.563	0.061
		7	0.136	0.245	11.571	0.072
		8	0.015	-0.114	11.584	0.115
		9	-0.126	-0.088	12.496	0.130
		10	-0.143	0.011	13.711	0.133
		11	0.016	-0.002	13.727	0.186
		12	0.127	-0.000	14.744	0.194
		13	0.167	0.198	16.569	0.167
		14	-0.019	-0.132	16.593	0.219
		15	-0.255	-0.205	21.141	0.098
		16	-0.201	0.028	24.056	0.064
		17	0.031	-0.006	24.130	0.087
		18	0.182	-0.027	26.715	0.062
		19	-0.047	-0.119	26.891	0.081
		20	-0.185	-0.127	29.789	0.055

Dependent Variable: PRECIO_INTERNACIOAL_VI_2024
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:03
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

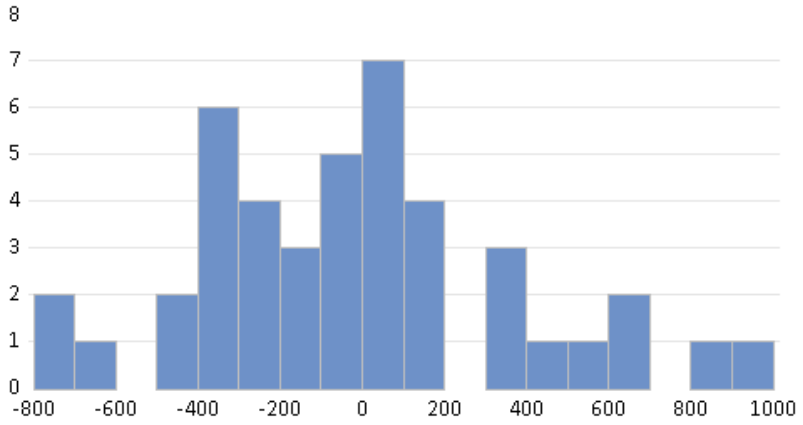
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	1.039179	0.148900	6.979024	0.0000
AR(2)	-0.285599	0.140693	-2.029946	0.0489

R-squared	0.681592	Mean dependent var	-59.47634
Adjusted R-squared	0.673826	S.D. dependent var	689.7585
S.E. of regression	393.9323	Akaike info criterion	14.83563
Sum squared resid	6362490.	Schwarz criterion	14.91755
Log likelihood	-316.9661	Hannan-Quinn criter.	14.86584
Durbin-Watson stat	1.732396		

Inverted AR Roots	.52-.13i	.52+.13i

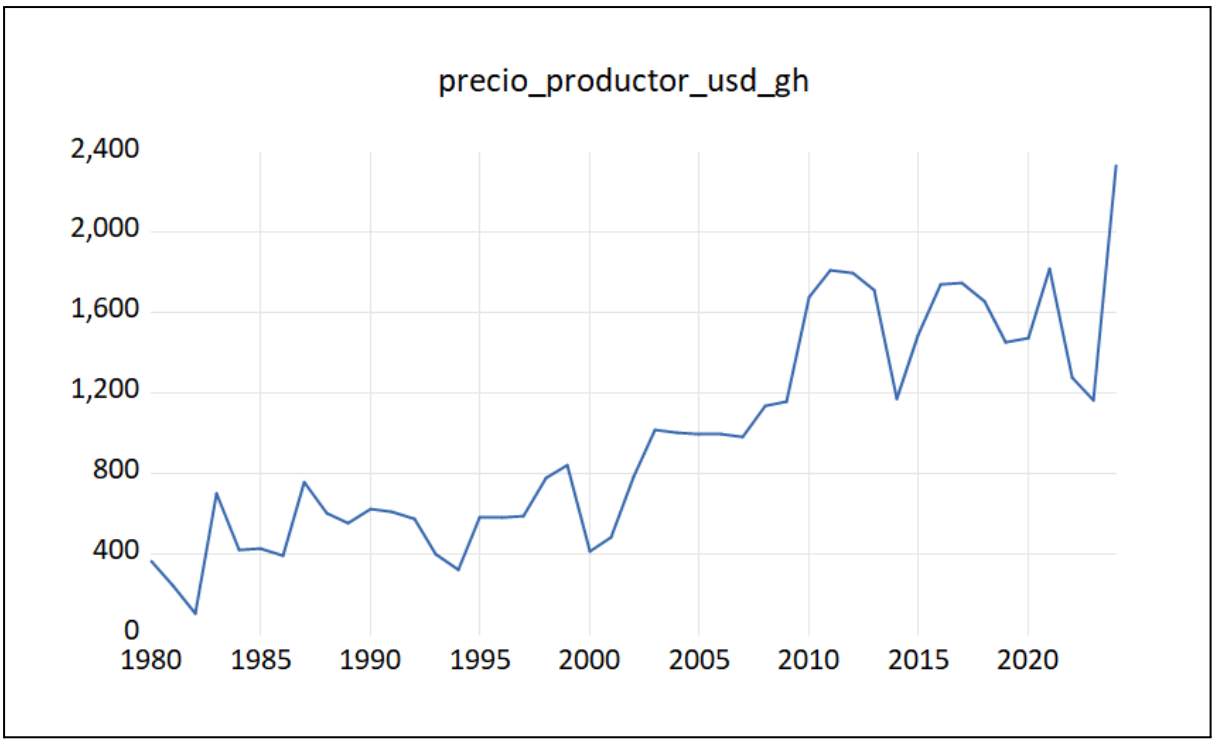
Date: 11/02/26 Time: 13:03
Sample (adjusted): 1982 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.098	0.098	0.4438	
		2	-0.145	-0.156	1.4328	
		3	-0.164	-0.137	2.7383	0.098
		4	0.089	0.103	3.1351	0.209
		5	0.115	0.057	3.8096	0.283
		6	-0.092	-0.115	4.2501	0.373
		7	0.109	0.195	4.8883	0.430
		8	0.009	-0.034	4.8929	0.558
		9	-0.107	-0.137	5.5411	0.594
		10	-0.153	-0.071	6.9223	0.545
		11	-0.018	-0.028	6.9430	0.643
		12	0.041	-0.072	7.0500	0.721
		13	0.139	0.182	8.2948	0.687
		14	-0.023	-0.047	8.3316	0.759
		15	-0.239	-0.248	12.266	0.506
		16	-0.178	-0.077	14.542	0.410
		17	-0.007	-0.031	14.546	0.485
		18	0.140	-0.014	16.053	0.449
		19	-0.088	-0.093	16.671	0.477
		20	-0.176	-0.165	19.269	0.375



Series: Residuals
Sample 1982 2024
Observations 43

Mean	-22.58094
Median	-25.69318
Maximum	930.7882
Minimum	-763.0330
Std. Dev.	388.5432
Skewness	0.427440
Kurtosis	3.012949
Jarque-Bera	1.309684
Probability	0.519524



En el gráfico se observa en un principio valores estables en el precio del productor, luego una fuerte subida entre los años 2000-2010, luego cierta estabilidad, y finalmente un gran salto en 2023-2024 que puede deberse a la escasez mundial en la oferta de cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO PRODUCTOR USD GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.985067	0.0165
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO PRODUCTOR USD GH)
Method: Least Squares
Date: 10/14/26 Time: 18:53
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO PRODUCTOR USD GH(-1)	-0.612121	0.153604	-3.985067	0.0003
C	94.54949	81.62358	1.158360	0.2534
@TREND("1980")	23.38542	6.061770	3.857853	0.0004

R-squared	0.287137	Mean dependent var	44.89464
Adjusted R-squared	0.252363	S.D. dependent var	291.8291
S.E. of regression	252.3330	Akaike info criterion	13.96512
Sum squared resid	2610549.	Schwarz criterion	14.08677
Log likelihood	-304.2327	Hannan-Quinn criter.	14.01024
F-statistic	8.257269	Durbin-Watson stat	1.682933
Prob(F-statistic)	0.000970		

La tendencia determinista resulta significativa, por lo que observo el p-valor asociado al estadístico tau, como 0.0165<0.05 rechazo la hipótesis nula de raíz unitario, por lo que no debo diferenciar la serie.

Date: 10/14/26 Time: 18:53
Sample: 1980 2024
Included observations: 45



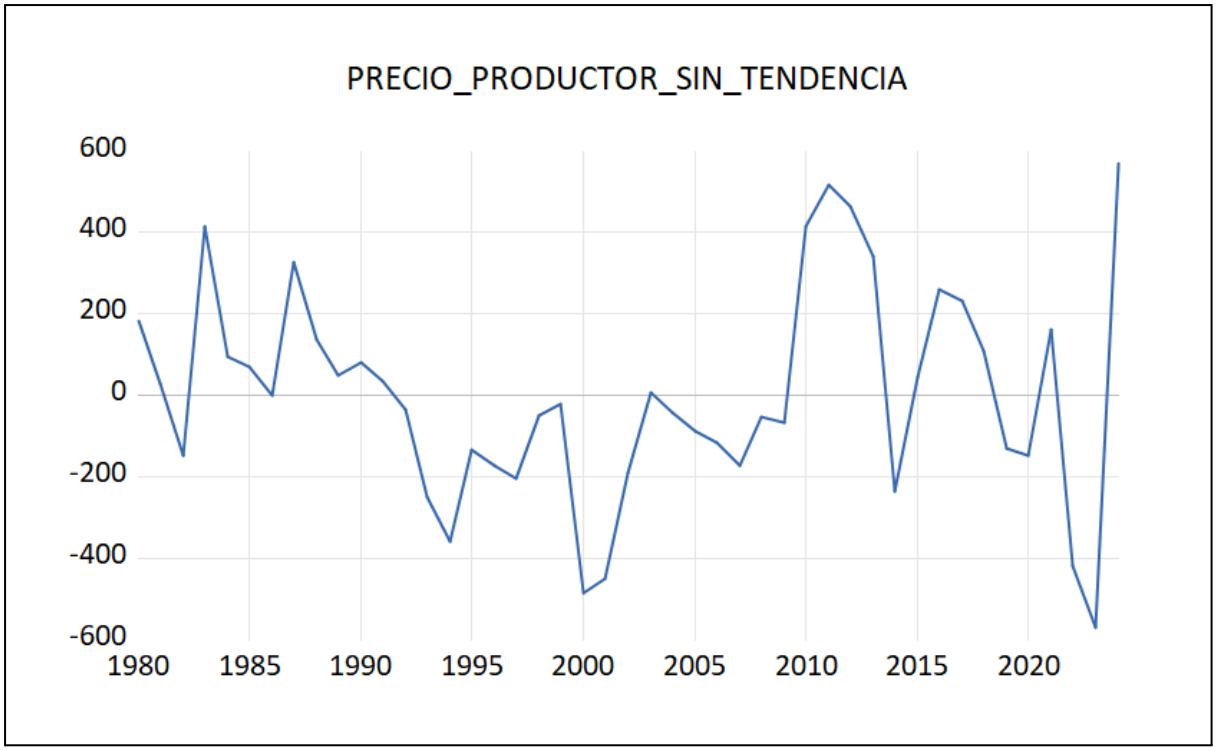
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.769	0.769	28.423	0.000
		2	0.692	0.246	51.956	0.000
		3	0.684	0.245	75.521	0.000
		4	0.635	0.054	96.332	0.000
		5	0.596	0.043	115.10	0.000
		6	0.559	0.005	132.06	0.000
		7	0.511	-0.033	146.58	0.000
		8	0.472	-0.019	159.28	0.000
		9	0.368	-0.206	167.24	0.000
		10	0.260	-0.222	171.31	0.000
		11	0.262	0.075	175.60	0.000
		12	0.224	0.014	178.81	0.000
		13	0.175	0.030	180.84	0.000
		14	0.103	-0.105	181.57	0.000
		15	-0.014	-0.243	181.58	0.000
		16	-0.073	-0.092	181.98	0.000
		17	-0.139	-0.088	183.43	0.000
		18	-0.153	0.130	185.27	0.000
		19	-0.171	0.033	187.65	0.000
		20	-0.190	0.046	190.71	0.000

Dependent Variable: PRECIO PRODUCTOR USD GH
Method: Least Squares
Date: 10/14/26 Time: 19:00
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
@TREND	36.11294	3.058563	11.80716	0.0000
C	177.2162	78.13815	2.267985	0.0284

R-squared	0.764266	Mean dependent var	971.7009
Adjusted R-squared	0.758784	S.D. dependent var	542.5442
S.E. of regression	266.4638	Akaike info criterion	14.05178
Sum squared resid	3053128.	Schwarz criterion	14.13208
Log likelihood	-314.1651	Hannan-Quinn criter.	14.08171
F-statistic	139.4090	Durbin-Watson stat	1.200557
Prob(F-statistic)	0.000000		

Como tanto la tendencia como el intercepto resultan significativos al 5%, obtengo la serie de los residuos para analizar la serie sin tendencia determinística:



Date: 10/14/26 Time: 19:01
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.341	0.341	5.5955	0.018
		2	0.072	-0.051	5.8488	0.054
		3	0.179	0.193	7.4603	0.059
		4	0.110	-0.015	8.0876	0.088
		5	0.080	0.061	8.4239	0.134
		6	0.070	-0.000	8.6864	0.192
		7	0.086	0.060	9.0988	0.246
		8	0.086	0.026	9.5238	0.300
		9	-0.132	-0.213	10.552	0.308
		10	-0.409	-0.384	20.645	0.024
		11	-0.302	-0.146	26.311	0.006
		12	-0.209	-0.124	29.117	0.004
		13	-0.111	0.114	29.929	0.005
		14	-0.035	0.112	30.013	0.008
		15	-0.150	-0.056	31.589	0.007
		16	-0.205	-0.098	34.652	0.004
		17	-0.279	-0.204	40.540	0.001
		18	-0.167	0.064	42.716	0.001
		19	-0.103	-0.086	43.583	0.001
		20	-0.054	-0.107	43.830	0.002

Se ve como cambia el correlograma con respecto al anterior, tanto la FAC como la FACP dejan de ser significativas en el segundo rezago

Dependent Variable: PRECIO_PRODUCTOR_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/20/26 Time: 18:11
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Failure to improve likelihood (non-zero gradients) after 19 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	-0.463842	0.167211	-2.773991	0.0082
MA(1)	0.921565	0.060336	15.27381	0.0000

R-squared	0.230940	Mean dependent var	-4.154996
Adjusted R-squared	0.212630	S.D. dependent var	264.9680
S.E. of regression	235.1164	Akaike info criterion	13.80243
Sum squared resid	2321749.	Schwarz criterion	13.88353
Log likelihood	-301.6534	Hannan-Quinn criter.	13.83250
Durbin-Watson stat	1.566188		

Inverted AR Roots	-.46
Inverted MA Roots	-.92

Date: 10/20/26 Time: 18:12
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.137	0.137	0.8855	
		2	0.072	0.055	1.1380	
		3	0.051	0.035	1.2646	0.261
		4	0.137	0.125	2.2154	0.330
		5	0.001	-0.038	2.2155	0.529
		6	0.079	0.071	2.5490	0.636
		7	-0.000	-0.027	2.5490	0.769
		8	0.115	0.100	3.2938	0.771
		9	-0.122	-0.156	4.1596	0.761
		10	-0.316	-0.330	10.098	0.258
		11	-0.276	-0.228	14.764	0.098
		12	-0.081	-0.050	15.179	0.126
		13	-0.094	-0.004	15.761	0.150
		14	0.093	0.237	16.350	0.176
		15	-0.163	-0.103	18.203	0.150
		16	-0.084	-0.037	18.719	0.176
		17	-0.256	-0.251	23.613	0.072
		18	-0.107	-0.061	24.496	0.079
		19	-0.086	-0.021	25.097	0.093
		20	0.023	-0.096	25.140	0.121

Ambos coeficientes son significativos y el residuo es ruido blanco, sin embargo la raíz del coeficiente MA(1) está muy cerca del círculo unitario.

Dependent Variable: PRECIO PRODUCTOR SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:02
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 10 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.381693	0.149479	2.553495	0.0143

R-squared	0.131451	Mean dependent var	-4.154996
Adjusted R-squared	0.131451	S.D. dependent var	264.9680
S.E. of regression	246.9395	Akaike info criterion	13.87863
Sum squared resid	2622101.	Schwarz criterion	13.91918
Log likelihood	-304.3298	Hannan-Quinn criter.	13.89367
Durbin-Watson stat	1.669582		

Inverted AR Roots	.38

Date: 10/14/26 Time: 19:06
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.047	0.047	0.1048	
		2	-0.158	-0.161	1.3103	0.252
		3	0.085	0.104	1.6686	0.434
		4	0.055	0.018	1.8201	0.611
		5	0.016	0.043	1.8333	0.766
		6	0.007	0.006	1.8359	0.871
		7	0.024	0.027	1.8686	0.931
		8	0.146	0.144	3.0640	0.879
		9	-0.012	-0.029	3.0729	0.930
		10	-0.345	-0.321	10.167	0.337
		11	-0.158	-0.181	11.705	0.305
		12	-0.078	-0.209	12.094	0.357
		13	-0.000	0.008	12.094	0.438
		14	0.117	0.169	13.021	0.446
		15	-0.056	0.019	13.240	0.508
		16	-0.078	-0.008	13.677	0.550
		17	-0.196	-0.259	16.566	0.414
		18	-0.065	-0.008	16.893	0.462
		19	-0.046	-0.076	17.067	0.518
		20	-0.012	-0.054	17.079	0.585

Nuevamente el coeficiente AR(1) es significativo, con raíz dentro del círculo unitario y con residuo ruido blanco, sin embargo debo probar como se comporta el coeficiente MA(1) sin el componente AR(1):

Dependent Variable: PRECIO PRODUCTOR SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:04
Sample: 1980 2024
Included observations: 45
Failure to improve likelihood (non-zero gradients) after 15 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1979

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.566374	0.136159	4.159653	0.0001

R-squared	0.181616	Mean dependent var	-1.09E-13
Adjusted R-squared	0.181616	S.D. dependent var	263.4184
S.E. of regression	238.3004	Akaike info criterion	13.80691
Sum squared resid	2498631.	Schwarz criterion	13.84706
Log likelihood	-309.6555	Hannan-Quinn criter.	13.82188
Durbin-Watson stat	1.841086		

Inverted MA Roots	-.57

Date: 10/14/26 Time: 19:05
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.029	-0.029	0.0398	
		2	-0.029	-0.030	0.0810	0.776
		3	0.091	0.089	0.4953	0.781
		4	0.078	0.083	0.8117	0.847
		5	0.011	0.022	0.8183	0.936
		6	0.025	0.023	0.8526	0.974
		7	0.044	0.032	0.9592	0.987
		8	0.088	0.084	1.4002	0.986
		9	0.008	0.010	1.4036	0.994
		10	-0.310	-0.323	7.2047	0.616
		11	-0.148	-0.222	8.5642	0.574
		12	-0.058	-0.135	8.7793	0.642
		13	-0.062	-0.042	9.0304	0.700
		14	0.109	0.214	9.8366	0.707
		15	-0.104	-0.015	10.600	0.717
		16	-0.055	-0.012	10.823	0.765
		17	-0.215	-0.256	14.314	0.575
		18	-0.049	-0.049	14.506	0.631
		19	-0.071	-0.021	14.914	0.668
		20	-0.035	-0.076	15.018	0.721

El coeficiente MA(1) por separado arroja el problema de “Failure to improve likelihood…”, lo cual puede deberse a la poca cantidad de observaciones, por lo que este modelo podría ser problemático.

Sin embargo si se pasa por alto eso y se quiere comparar ambos modelos, necesito comparar con el criterio de información Akaike, el cual requiere que ambos modelos tengan la misma cantidad de observaciones, por lo que estimo nuevamente el MA(1) con 44 observaciones:

Dependent Variable: PRECIO PRODUCTOR SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:06
Sample: 1981 2024
Included observations: 44
Failure to improve likelihood (non-zero gradients) after 22 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.603754	0.122461	4.930154	0.0000

R-squared	0.179885	Mean dependent var	-4.154996
Adjusted R-squared	0.179885	S.D. dependent var	264.9680
S.E. of regression	239.9555	Akaike info criterion	13.82125
Sum squared resid	2475882.	Schwarz criterion	13.86180
Log likelihood	-303.0675	Hannan-Quinn criter.	13.83629
Durbin-Watson stat	1.903025		

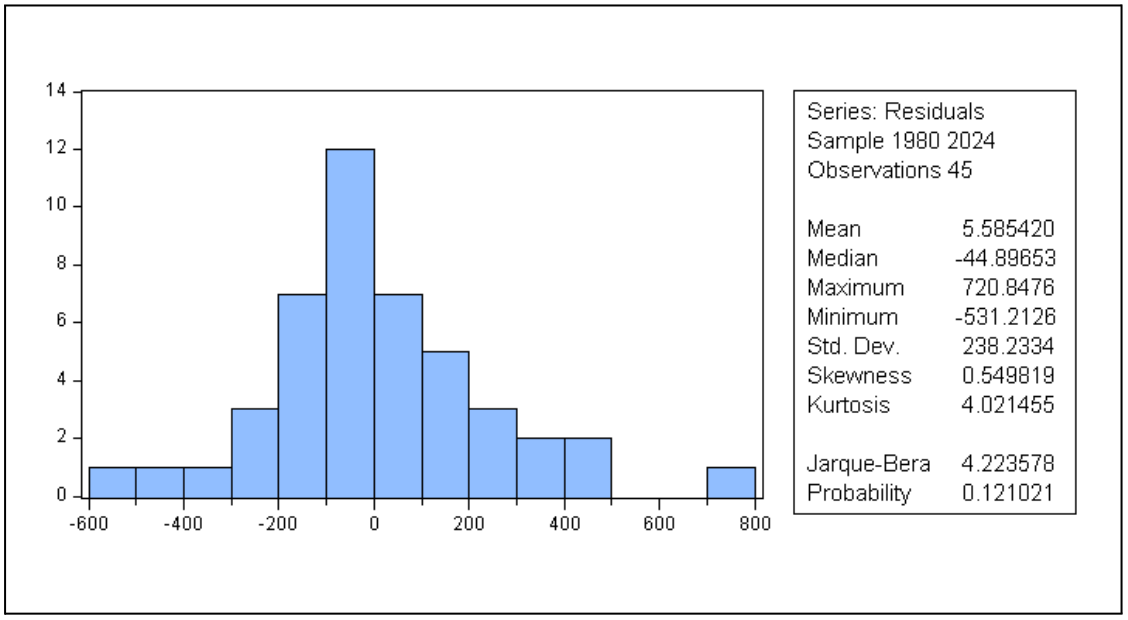
Inverted MA Roots	-.60

Date: 10/14/26 Time: 19:07
Sample: 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.056	-0.056	0.1453	
		2	-0.005	-0.008	0.1466	0.702
		3	0.042	0.041	0.2321	0.890
		4	0.099	0.105	0.7325	0.866
		5	-0.010	0.002	0.7378	0.947
		6	0.035	0.035	0.8042	0.977
		7	0.014	0.010	0.8155	0.992
		8	0.091	0.085	1.2857	0.989
		9	0.004	0.013	1.2868	0.996
		10	-0.304	-0.318	6.8032	0.658
		11	-0.147	-0.222	8.1347	0.616
		12	-0.046	-0.116	8.2683	0.689
		13	-0.054	-0.052	8.4560	0.749
		14	0.132	0.236	9.6353	0.723
		15	-0.111	-0.022	10.497	0.725
		16	-0.034	-0.004	10.580	0.782
		17	-0.211	-0.252	13.921	0.605
		18	-0.043	-0.084	14.065	0.663
		19	-0.072	-0.020	14.486	0.697
		20	-0.002	-0.080	14.486	0.755

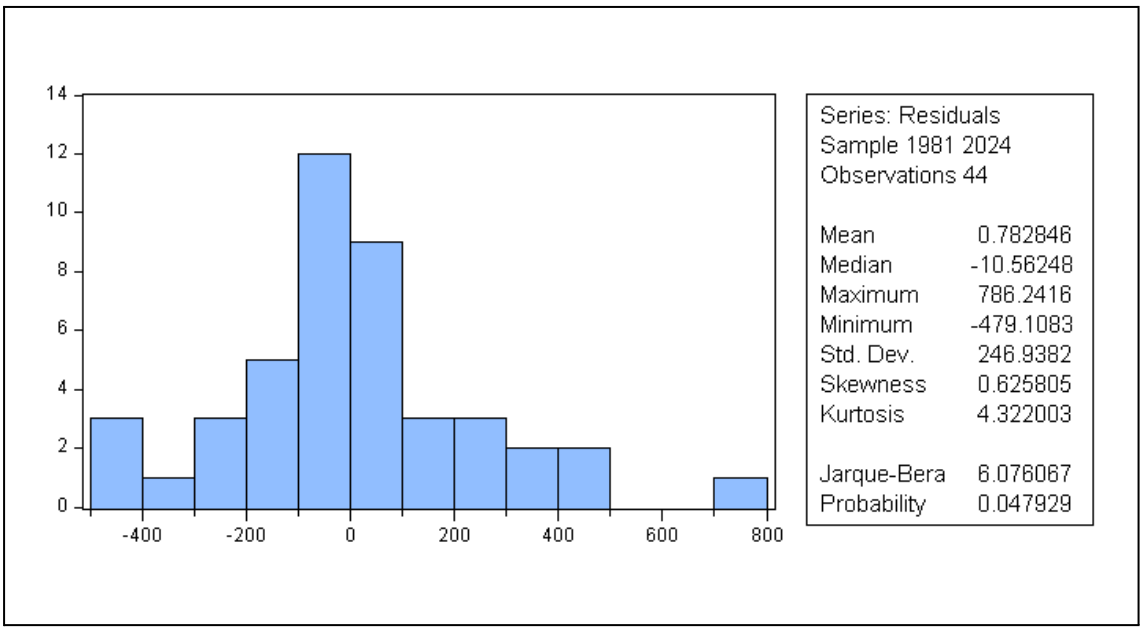
La nueva estimación del MA(1) sigue arrojando el problema de convergencia, sin embargo tiene un mejor AIC que el AR(1), 13.82<13.87.
Por lo que si ignoramos el problema de la falla de convergencia, el modelo elegido sería el MA(1), de lo contrario el modelo sería el AR(1)

Jarque-Bera test MA(1)



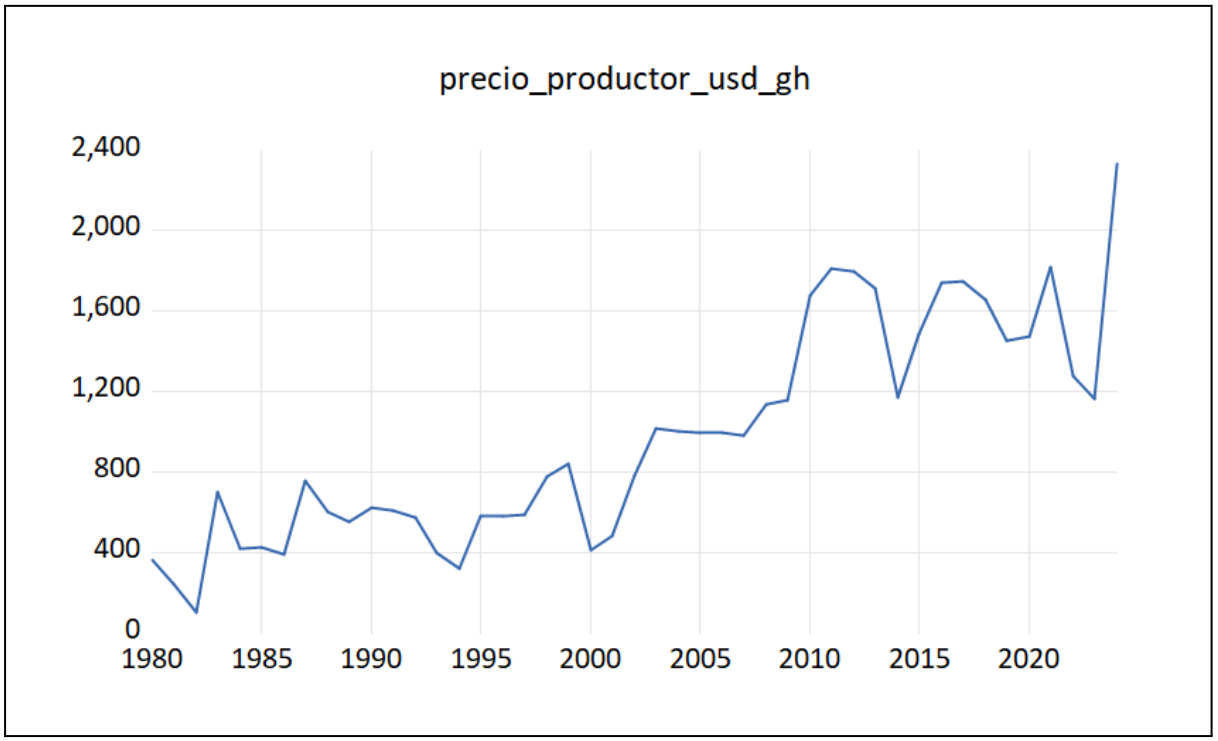
No se rechaza la hipótesis nula, por lo que los residuos del MA(1) se comportan normalmente.

Jarque-Bera test AR(1)



Se rechaza por muy poco (0.047929<0.05) el test de normalidad de residuos de Jarque-Bera con un nivel de significancia del 5%.

Agregando una Variable de Impacto:



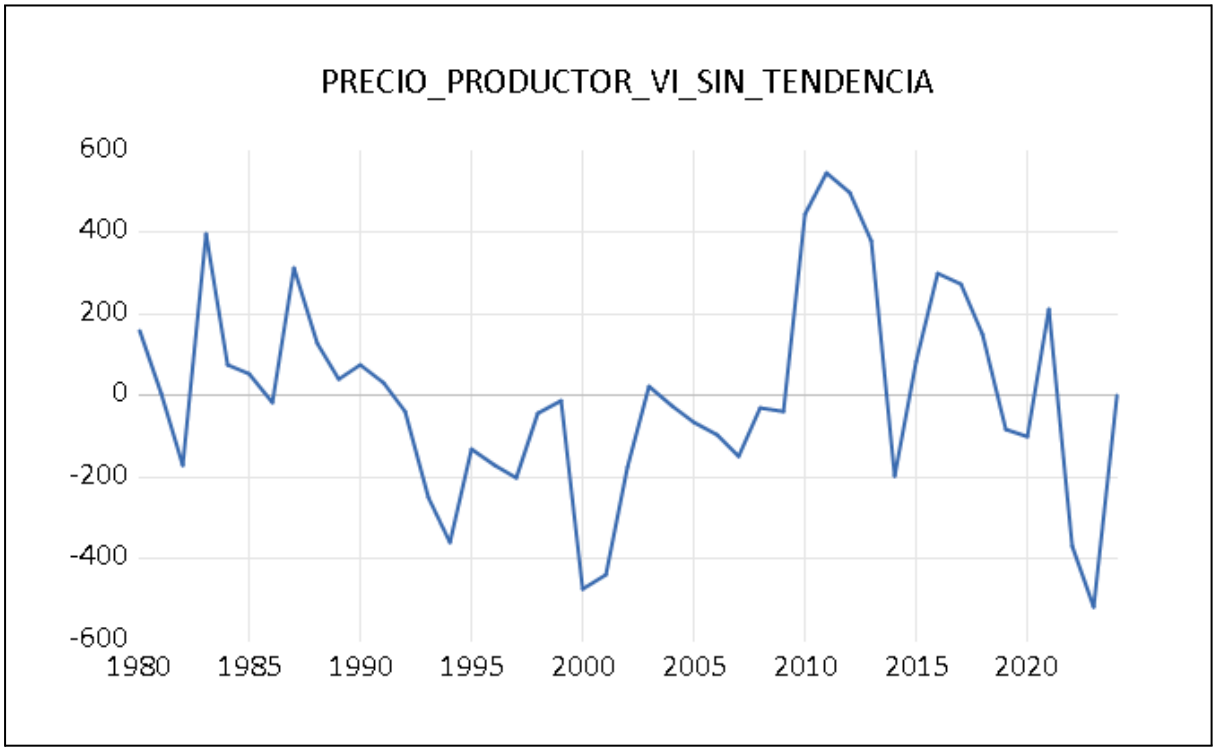
Durante los últimos años está ocurriendo una crisis respecto a la producción del cacao en África Occidental, provocada por un virus sobre la planta de cacao, que afectó a Ghana y Costa de Marfil, dos de los mayores productores de cacao a nivel mundial (60% de la oferta mundial).
Es por eso que decidí incorporar una variable de intervencion que refleje ese shock tanto en la producción como en el precio internacional, y por lo tanto en el precio del productor.
Como se puede ver en el primer gráfico de la serie, esto se vio reflejado en la serie con un gran salto durante el anio 2024, debido a que el gobierno Ghanes debió reajustar el precio del productor a mitad de anio debido a la gran diferencia que había con el precio internacional.

Dependent Variable: PRECIO_PRODUCTOR_USD_GH
Method: Least Squares
Date: 11/02/26 Time: 13:05
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	203.0896	75.14324	2.702699	0.0099
@TREND	34.30782	3.009343	11.40044	0.0000
VI_2024	622.7663	265.1383	2.348836	0.0236

R-squared	0.791636	Mean dependent var	971.7009
Adjusted R-squared	0.781714	S.D. dependent var	542.5442
S.E. of regression	253.4824	Akaike info criterion	13.97281
Sum squared resid	2698641.	Schwarz criterion	14.09325
Log likelihood	-311.3881	Hannan-Quinn criter.	14.01771
F-statistic	79.78529	Durbin-Watson stat	0.976569
Prob(F-statistic)	0.000000		

La variable de intervención (toma valor 0 en todos los años y 1 en 2024) resultó significativa junto a la tendencia y el intercepto



Null Hypothesis: PRECIO_PRODUCTOR_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.773091	0.0004
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_PRODUCTOR_VI_SIN_TENDENCIA)
Method: Least Squares
Date: 11/02/26 Time: 13:07
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO_PRODUCTOR_VI_SIN_TENDENC	-0.492848	0.130622	-3.773091	0.0005

R-squared	0.248568	Mean dependent var	-3.566965
Adjusted R-squared	0.248568	S.D. dependent var	247.5390
S.E. of regression	214.5797	Akaike info criterion	13.59770
Sum squared resid	1979911.	Schwarz criterion	13.63825
Log likelihood	-298.1495	Hannan-Quinn criter.	13.61274
Durbin-Watson stat	1.841784		

Luego de quitar la tendencia e intercepto a la serie, y quitar el impacto del año 2024 a la serie, verificamos que el test de dickey fuller rechaza la presencia de raíz unitaria.

Date: 11/03/26 Time: 09:50
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.507	0.507	12.363	0.000
		2	0.179	-0.105	13.937	0.001
		3	0.173	0.172	15.452	0.001
		4	0.168	0.023	16.916	0.002
		5	0.134	0.043	17.859	0.003
		6	0.071	-0.031	18.133	0.006
		7	0.059	0.032	18.327	0.011
		8	0.052	-0.010	18.481	0.018
		9	-0.149	-0.255	19.780	0.019
		10	-0.397	-0.319	29.306	0.001
		11	-0.403	-0.149	39.417	0.000
		12	-0.332	-0.146	46.500	0.000
		13	-0.242	0.013	50.383	0.000
		14	-0.147	0.097	51.849	0.000
		15	-0.177	-0.032	54.064	0.000
		16	-0.244	-0.080	58.390	0.000
		17	-0.301	-0.142	65.251	0.000
		18	-0.186	0.093	67.973	0.000
		19	-0.121	-0.123	69.160	0.000
		20	-0.074	-0.083	69.627	0.000

Dependent Variable: PRECIO_PRODUCTOR_VI_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:11
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.507152	0.130622	3.882592	0.0004

R-squared	0.259417	Mean dependent var	-3.566965
Adjusted R-squared	0.259417	S.D. dependent var	249.3456
S.E. of regression	214.5797	Akaike info criterion	13.59770
Sum squared resid	1979911.	Schwarz criterion	13.63825
Log likelihood	-298.1495	Hannan-Quinn criter.	13.61274
Durbin-Watson stat	1.841784		

Inverted AR Roots .51

Date: 11/02/26 Time: 13:13
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.060	0.060	0.1675	
		2	-0.150	-0.154	1.2525	0.263
		3	0.019	0.040	1.2709	0.530
		4	0.086	0.061	1.6475	0.649
		5	0.062	0.063	1.8485	0.764
		6	-0.008	0.005	1.8516	0.869
		7	-0.009	0.005	1.8563	0.932
		8	0.153	0.149	3.1707	0.869
		9	-0.015	-0.047	3.1837	0.922
		10	-0.304	-0.276	8.6978	0.466
		11	-0.186	-0.190	10.830	0.371
		12	-0.120	-0.236	11.738	0.384
		13	-0.066	-0.146	12.026	0.444
		14	0.057	0.060	12.242	0.508
		15	-0.040	0.020	12.352	0.578
		16	-0.080	0.004	12.813	0.617
		17	-0.212	-0.197	16.170	0.441
		18	-0.034	0.042	16.259	0.506
		19	-0.029	-0.070	16.325	0.570
		20	-0.010	-0.055	16.333	0.635

Dependent Variable: PRECIO_PRODUCTOR_VI_SIN_TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/02/26 Time: 13:12
Sample: 1980 2024
Included observations: 45
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.536318	0.132053	4.061395	0.0002
SIGMASQ	43937.87	9089.990	4.833654	0.0000

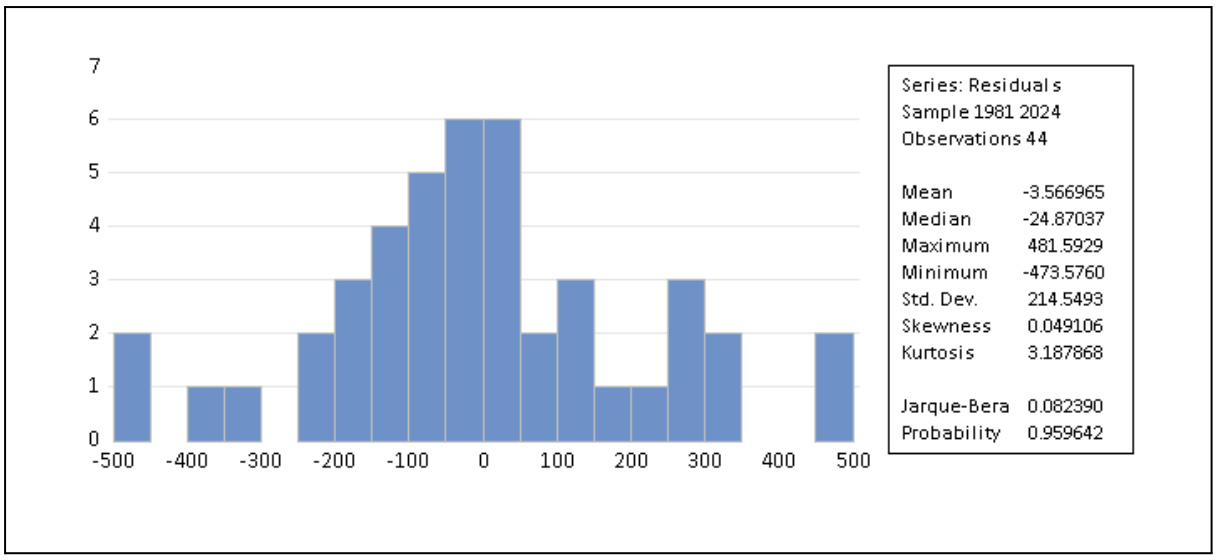
R-squared	0.267333	Mean dependent var	2.69E-13
Adjusted R-squared	0.250294	S.D. dependent var	247.6545
S.E. of regression	214.4330	Akaike info criterion	13.62483
Sum squared resid	1977204.	Schwarz criterion	13.70513
Log likelihood	-304.5588	Hannan-Quinn criter.	13.65477
Durbin-Watson stat	1.838321		

Inverted MA Roots -.54

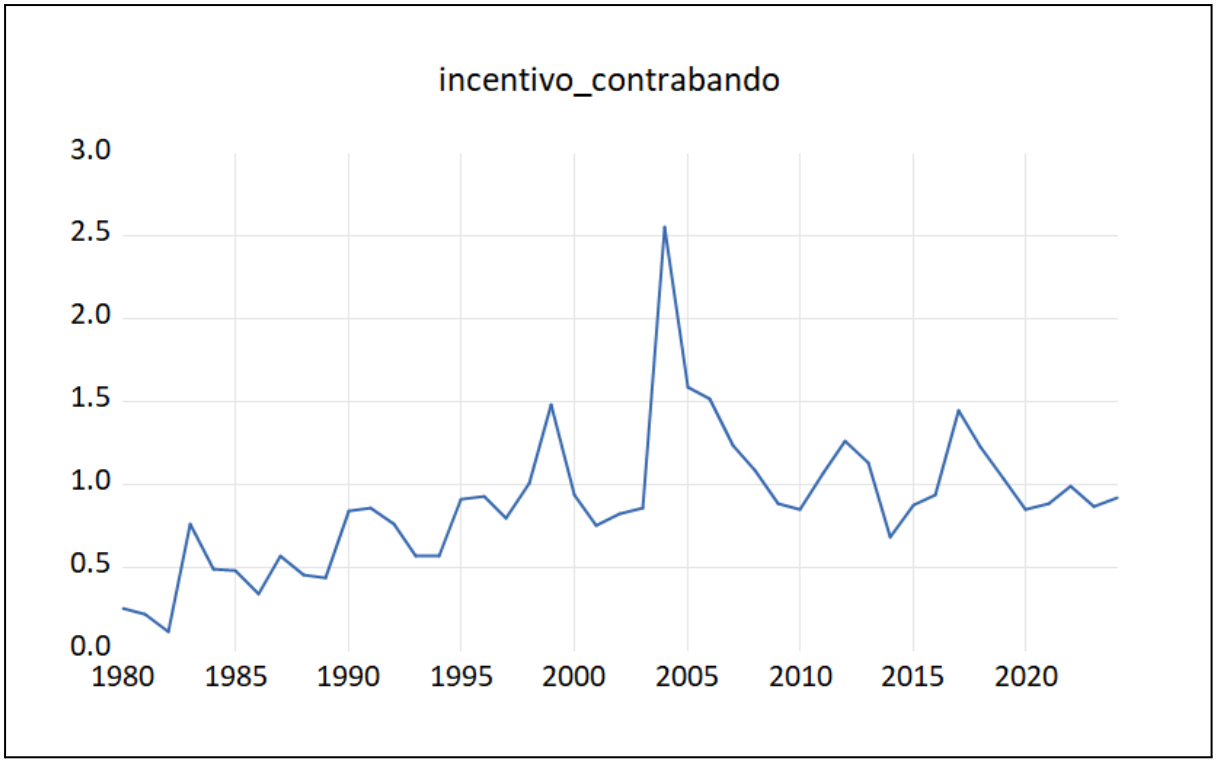
Date: 11/02/26 Time: 13:12
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.071	0.071	0.2454	
		2	0.119	0.115	0.9477	0.330
		3	0.068	0.053	1.1792	0.555
		4	0.127	0.108	2.0114	0.570
		5	0.069	0.043	2.2608	0.688
		6	0.039	0.005	2.3444	0.800
		7	0.015	-0.012	2.3573	0.884
		8	0.080	0.059	2.7269	0.909
		9	-0.059	-0.084	2.9325	0.939
		10	-0.295	-0.324	8.2034	0.514
		11	-0.236	-0.243	11.673	0.308
		12	-0.177	-0.156	13.671	0.252
		13	-0.160	-0.119	15.373	0.222
		14	-0.018	0.110	15.395	0.283
		15	-0.124	0.027	16.480	0.285
		16	-0.101	-0.001	17.231	0.305
		17	-0.252	-0.195	22.021	0.143
		18	-0.057	0.018	22.277	0.174
		19	-0.093	-0.023	22.976	0.192
		20	-0.037	-0.095	23.092	0.233

Comparando los modelos, ambos resultan significativos, con ruido blanco y raíces dentro del círculo unitario, por lo que los comparo con el criterio de información de Akaike. Según el cual elijo el modelo **AR(1)** (13.59<13.62)



Con el nuevo modelo incorporando la variable de intervención pude obtener un modelo con menor varianza, menor criterio de Akaike y con residuos ruido blanco, en comparación con el mejor modelo sin variable de intervención.



Se nota una leve tendencia entre 1980 y principios de los 2000 que termina con la presencia de un salto abrupto en el año 2004 que puede deberse a la guerra civil ocurrida en Costa de Marfil, que ocurrió entre los años 2002-2007, pero que tuvo su apogeo en el 2004. A partir del 2010 se observa cierta estabilidad.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: INCENTIVO CONTRABANDO has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.941365	0.0184
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO CONTRABANDO)
Method: Least Squares
Date: 10/14/26 Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO CONTRABANDO(-1)	-0.553339	0.140393	-3.941365	0.0003
C	0.332058	0.123311	2.692847	0.0102
@TREND("1980")	0.007811	0.004657	1.677212	0.1011

R-squared	0.277855	Mean dependent var	0.015094
Adjusted R-squared	0.242628	S.D. dependent var	0.381944
S.E. of regression	0.332394	Akaike info criterion	0.700757
Sum squared resid	4.529927	Schwarz criterion	0.822406
Log likelihood	-12.41665	Hannan-Quinn criter.	0.745870
F-statistic	7.887653	Durbin-Watson stat	2.020398
Prob(F-statistic)	0.001264		

Como la tendencia determinística no resulta significativa (p-valor = 0.1011), vuelvo a hacer el test con intercepto.

Null Hypothesis: INCENTIVO CONTRABANDO has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.525035	0.0118
Test critical values:	1% level	-3.588509	
	5% level	-2.929734	
	10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

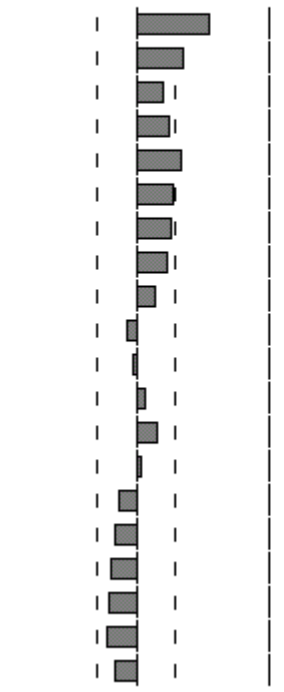
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO CONTRABANDO)
Method: Least Squares
Date: 10/14/26 Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO CONTRABANDO(-1)	-0.428289	0.121499	-3.525035	0.0010
C	0.396459	0.119683	3.312581	0.0019

R-squared	0.228308	Mean dependent var	0.015094
Adjusted R-squared	0.209935	S.D. dependent var	0.381944
S.E. of regression	0.339493	Akaike info criterion	0.721662
Sum squared resid	4.840729	Schwarz criterion	0.802761
Log likelihood	-13.87656	Hannan-Quinn criter.	0.751737
F-statistic	12.42587	Durbin-Watson stat	2.150524
Prob(F-statistic)	0.001038		

El intercepto resulta significativo por lo que analizo el p-valor asociado al estadístico tau, como 0.0118<0.05 rechazo la hipótesis nula de raíz unitaria, por lo que proceso a analizar el correlograma de la serie en niveles:

Date: 10/14/26 Time: 19:11
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.572	0.572	15.706	0.000
		2	0.373	0.069	22.546	0.000
		3	0.211	-0.039	24.795	0.000
		4	0.266	0.212	28.439	0.000
		5	0.350	0.198	34.927	0.000
		6	0.289	-0.049	39.459	0.000
		7	0.274	0.095	43.653	0.000
		8	0.251	0.088	47.244	0.000
		9	0.144	-0.162	48.462	0.000
		10	-0.075	-0.331	48.806	0.000
		11	-0.025	0.169	48.846	0.000
		12	0.071	0.109	49.166	0.000
		13	0.156	-0.065	50.780	0.000
		14	0.048	-0.097	50.935	0.000
		15	-0.126	-0.061	52.054	0.000
		16	-0.166	-0.071	54.060	0.000
		17	-0.190	-0.131	56.798	0.000
		18	-0.218	-0.090	60.527	0.000
		19	-0.226	-0.021	64.693	0.000
		20	-0.171	-0.035	67.165	0.000

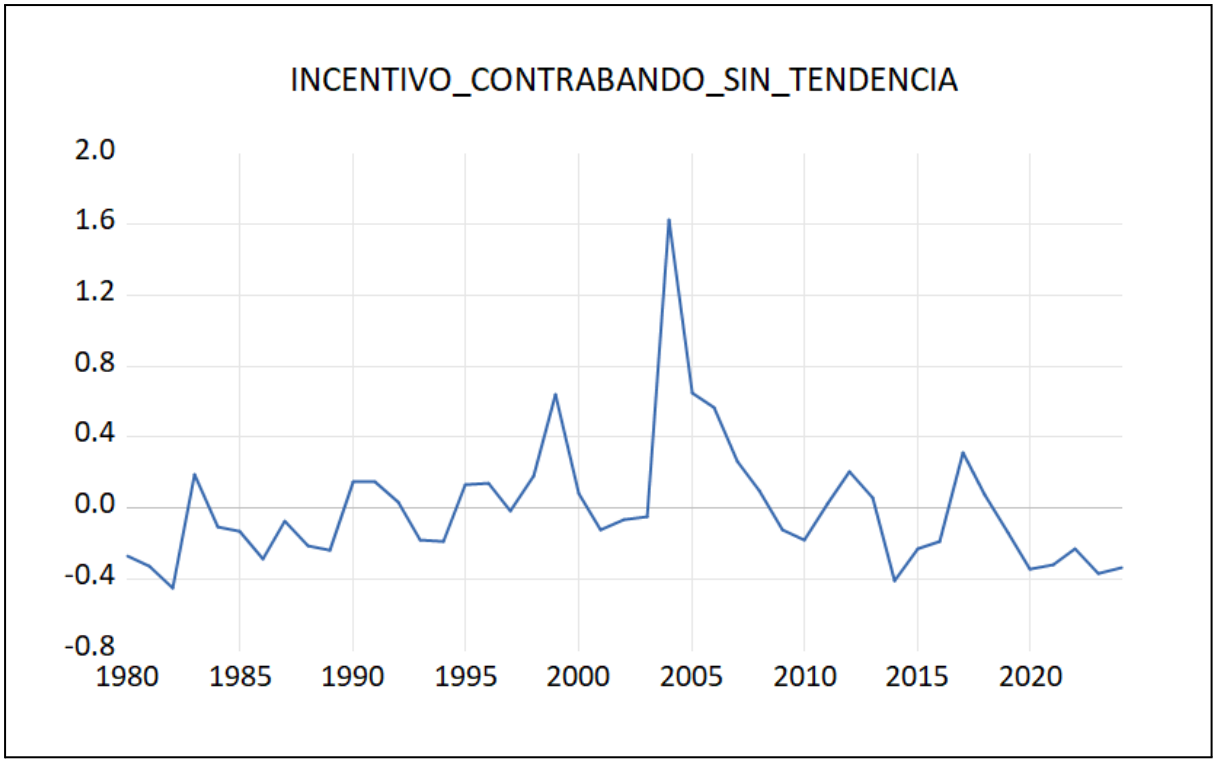
Dependent Variable: INCENTIVO CONTRABANDO
Method: Least Squares
Date: 10/14/26 Time: 19:12
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
@TREND	0.016547	0.004190	3.949091	0.0003
C	0.527008	0.107046	4.923183	0.0000

R-squared	0.266153	Mean dependent var	0.891044
Adjusted R-squared	0.249087	S.D. dependent var	0.421261
S.E. of regression	0.365045	Akaike info criterion	0.865833
Sum squared resid	5.730078	Schwarz criterion	0.946129
Log likelihood	-17.48123	Hannan-Quinn criter.	0.895766
F-statistic	15.59532	Durbin-Watson stat	1.094744
Prob(F-statistic)	0.000286		

Notar que en el ADF la tendencia no resultaba significativa, sin embargo al estimarla junto al intercepto tiene un p-valor = 0.0003.

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:



Date: 10/14/26 Time: 19:23
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.436	0.436	9.1448	0.002
		2	0.186	-0.005	10.854	0.004
		3	0.005	-0.093	10.855	0.013
		4	0.070	0.126	11.105	0.025
		5	0.192	0.168	13.048	0.023
		6	0.130	-0.050	13.958	0.030
		7	0.155	0.110	15.289	0.032
		8	0.170	0.123	16.937	0.031
		9	0.049	-0.130	17.077	0.048
		10	-0.227	-0.338	20.199	0.027
		11	-0.174	0.093	22.082	0.024
		12	-0.030	0.094	22.140	0.036
		13	0.117	-0.015	23.048	0.041
		14	0.002	-0.128	23.048	0.059
		15	-0.215	-0.136	26.320	0.035
		16	-0.269	-0.114	31.598	0.011
		17	-0.288	-0.144	37.884	0.003
		18	-0.293	-0.122	44.591	0.000
		19	-0.263	-0.057	50.210	0.000
		20	-0.171	-0.085	52.686	0.000

Analizando el correlograma, tanto la FAC como la FACP resultan significativas solamente en el primer rezago (y el rezago 10 de la FACP)

Dependent Variable: INCENTIVO_CONTRABANDO_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/20/26 Time: 18:28
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Failure to improve likelihood (non-zero gradients) after 8 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.491736	0.295234	1.665583	0.1032
MA(1)	-0.060339	0.338114	-0.178459	0.8592

R-squared	0.197069	Mean dependent var	0.006215
Adjusted R-squared	0.177951	S.D. dependent var	0.362601
S.E. of regression	0.328759	Akaike info criterion	0.657404
Sum squared resid	4.539457	Schwarz criterion	0.738504
Log likelihood	-12.46289	Hannan-Quinn criter.	0.687480
Durbin-Watson stat	1.982554		

Inverted AR Roots	.49
Inverted MA Roots	.06

Si se estima un ARMA(1,1) ninguno de los dos coeficientes resulta significativo.

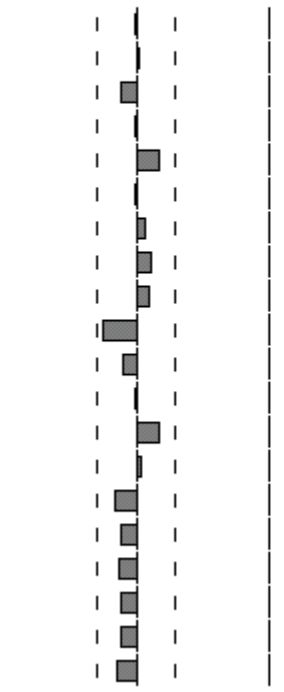
Dependent Variable: INCENTIVO_CONTRABANDO_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:26
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 3 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.445011	0.137157	3.244539	0.0023

R-squared	0.196426	Mean dependent var	0.006215
Adjusted R-squared	0.196426	S.D. dependent var	0.362601
S.E. of regression	0.325043	Akaike info criterion	0.612749
Sum squared resid	4.543090	Schwarz criterion	0.653299
Log likelihood	-12.48049	Hannan-Quinn criter.	0.627787
Durbin-Watson stat	2.011150		

Inverted AR Roots	.45

Date: 10/14/26 Time: 19:27
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term



	AC	PAC	Q-Stat	Prob
1	-0.014	-0.014	0.0089	
2	0.024	0.024	0.0369	0.848
3	-0.111	-0.110	0.6450	0.724
4	-0.012	-0.015	0.6520	0.884
5	0.181	0.189	2.3583	0.670
6	-0.008	-0.016	2.3617	0.797
7	0.070	0.057	2.6294	0.854
8	0.118	0.170	3.4088	0.845
9	0.099	0.107	3.9747	0.859
10	-0.257	-0.309	7.9165	0.543
11	-0.100	-0.089	8.5357	0.577
12	-0.015	0.024	8.5507	0.663
13	0.173	0.077	10.514	0.571
14	0.046	-0.032	10.658	0.639
15	-0.159	-0.089	12.431	0.572
16	-0.115	-0.101	13.385	0.573
17	-0.126	-0.132	14.574	0.556
18	-0.123	-0.153	15.747	0.542
19	-0.113	-0.073	16.782	0.538
20	-0.141	-0.218	18.454	0.492

Estimando solamente el AR(1) arroja coeficiente significativo, residuos ruido blanco y raíz dentro del círculo unitario, sin embargo debo probar como se comporta el componente MA(1) sin el AR(1):

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:28
Sample: 1980 2024
Included observations: 45
Failure to improve likelihood (non-zero gradients) after 12 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1979

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.368432	0.139890	2.633723	0.0116

R-squared	0.160268	Mean dependent var	1.78E-16
Adjusted R-squared	0.160268	S.D. dependent var	0.360873
S.E. of regression	0.330692	Akaike info criterion	0.646716
Sum squared resid	4.811730	Schwarz criterion	0.686864
Log likelihood	-13.55110	Hannan-Quinn criter.	0.661682
Durbin-Watson stat	1.833159		

Inverted MA Roots	-.37

Date: 10/14/26 Time: 19:28
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.072	0.072	0.2512	
		2	0.197	0.193	2.1649	0.141
		3	-0.068	-0.097	2.3957	0.302
		4	0.041	0.015	2.4845	0.478
		5	0.179	0.218	4.1759	0.383
		6	0.042	-0.006	4.2711	0.511
		7	0.114	0.039	4.9921	0.545
		8	0.105	0.139	5.6178	0.585
		9	0.096	0.050	6.1643	0.629
		10	-0.238	-0.358	9.5844	0.385
		11	-0.082	-0.063	10.007	0.440
		12	-0.047	0.107	10.146	0.517
		13	0.130	0.046	11.256	0.507
		14	0.016	-0.082	11.272	0.588
		15	-0.174	-0.129	13.416	0.494
		16	-0.161	-0.116	15.314	0.429
		17	-0.187	-0.133	17.962	0.326
		18	-0.196	-0.164	20.980	0.227
		19	-0.160	-0.042	23.062	0.188
		20	-0.173	-0.167	25.599	0.142

El coeficiente MA(1) por separado arroja el problema de “Failure to improve likelihood…”, lo cual puede deberse a la poca cantidad de observaciones, por lo que este modelo podría ser problemático.

Sin embargo si se pasa por alto eso y se quiere comparar ambos modelos, necesito comparar con el criterio de información Akaike, el cual requiere que ambos modelos tengan la misma cantidad de observaciones, por lo que estimo nuevamente el MA(1) con 44 observaciones:

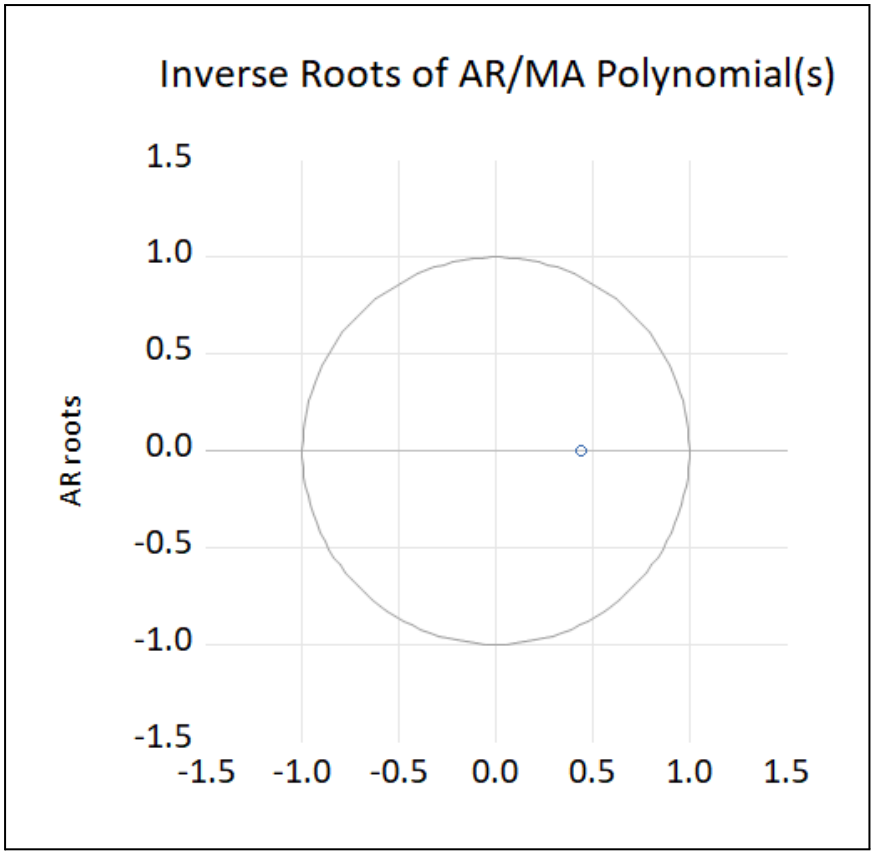
Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/14/26 Time: 19:29
Sample: 1981 2024
Included observations: 44
Failure to improve likelihood (non-zero gradients) after 12 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1980

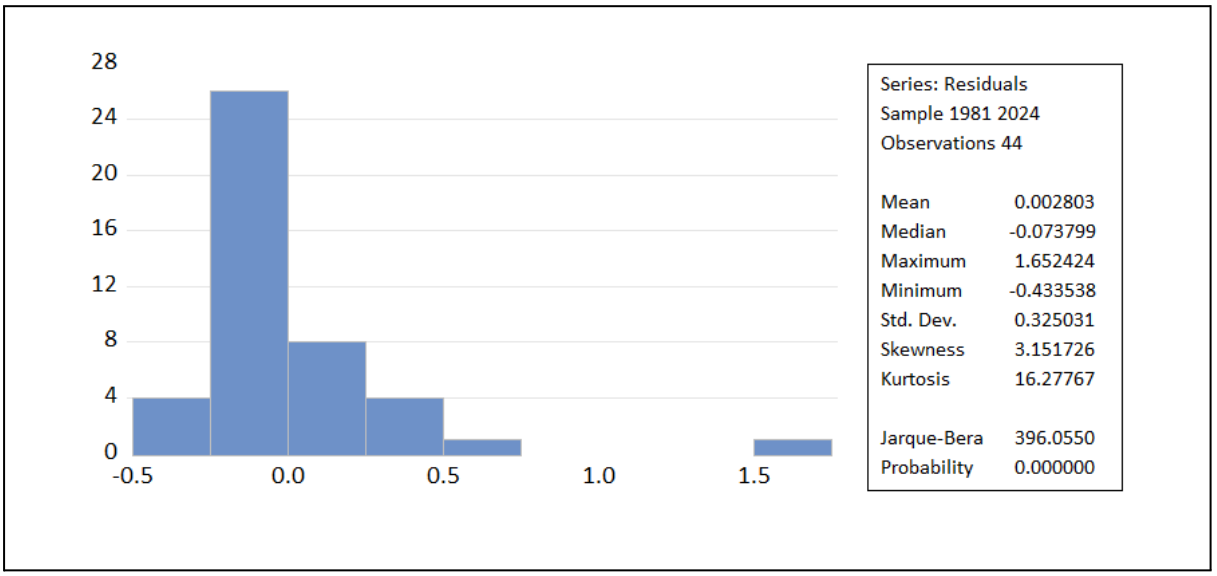
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.367926	0.142016	2.590728	0.0130

R-squared	0.157091	Mean dependent var	0.006215
Adjusted R-squared	0.157091	S.D. dependent var	0.362601
S.E. of regression	0.332904	Akaike info criterion	0.660539
Sum squared resid	4.765476	Schwarz criterion	0.701089
Log likelihood	-13.53187	Hannan-Quinn criter.	0.675577
Durbin-Watson stat	1.832781		

Inverted MA Roots	-.37

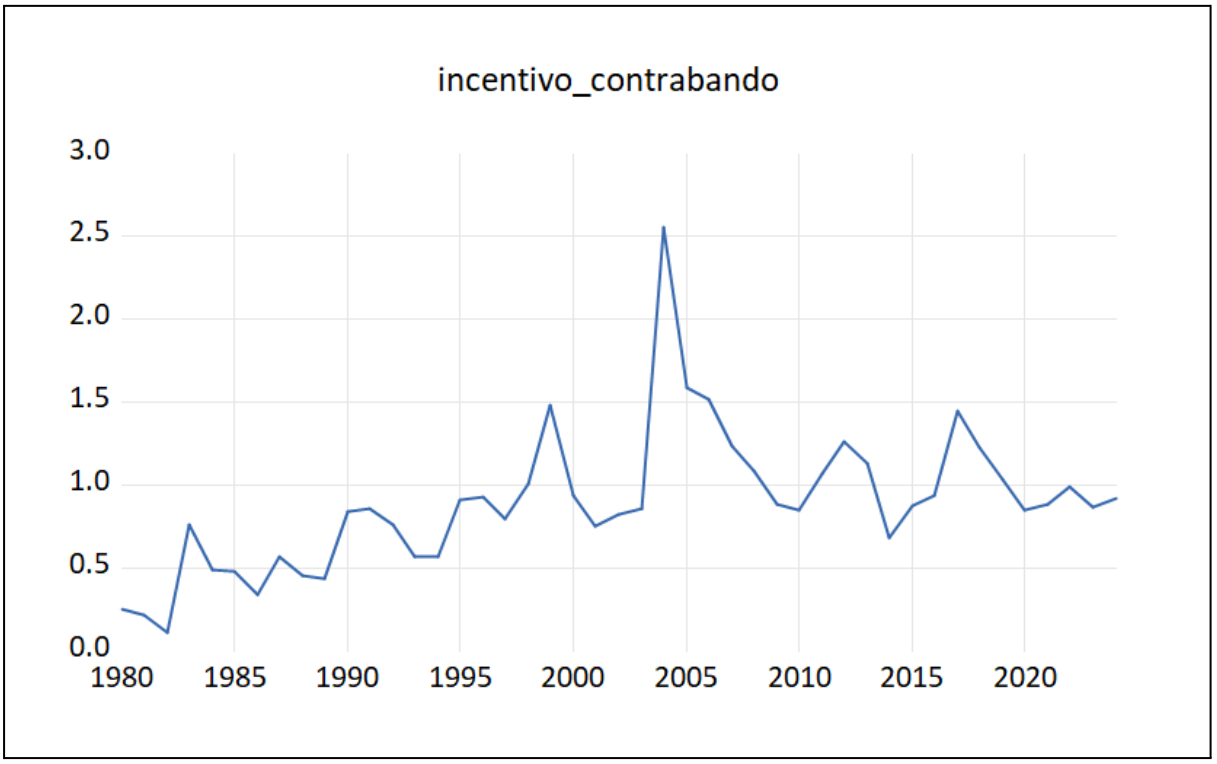
La nueva estimación del MA(1) sigue arrojando el problema de convergencia, y además tiene un peor AIC que el AR(1), 0.66>0.61, por lo que siguiendo este criterio debería quedarme con el modelo AR(1).





El test de Jarque-Bera del modelo AR(1) rechaza la hipótesis nula de que los residuos se comportan normalmente, por una fuerte asimetría positiva (3,151), y una curtosis elevada (16.27), **lo cual puede deberse al valor atípico del año 2004.**

Incorporando una Variable de intervención:



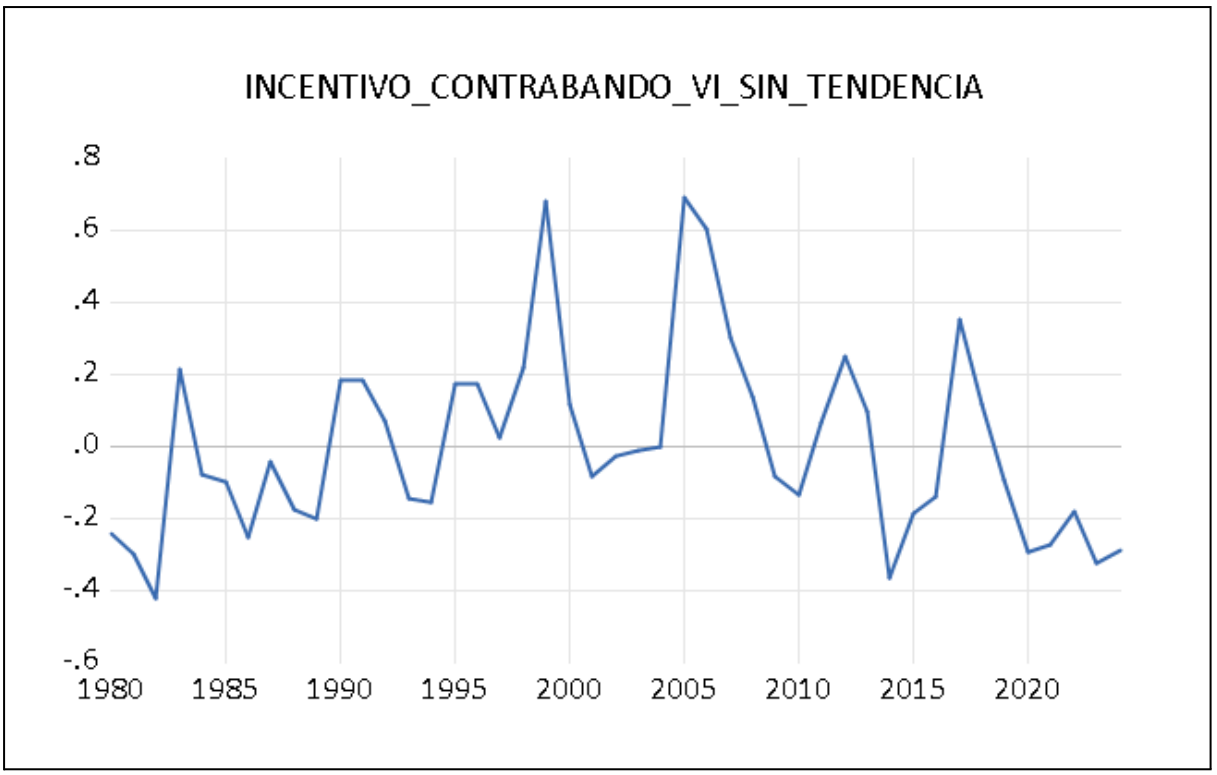
En este caso la variable de intervención se incorpora en el año 2004, debido a que fue el pico de la guerra civil de costa de marfil, que duró desde 2002 hasta 2007.
En ese año el precio de productor en costa de marfil fue extremadamente bajo, comparado con el de Ghana, lo que provoca que el incentivo de contrabando sea tan elevado.

Dependent Variable: INCENTIVO_CONTRABANDO
Method: Least Squares
Date: 11/02/26 Time: 13:16
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.499616	0.078658	6.351742	0.0000
@TREND	0.016108	0.003075	5.238586	0.0000
VI_2004	1.667653	0.270907	6.155822	0.0000

R-squared	0.614220	Mean dependent var	0.891044
Adjusted R-squared	0.595849	S.D. dependent var	0.421261
S.E. of regression	0.267808	Akaike info criterion	0.267244
Sum squared resid	3.012277	Schwarz criterion	0.387688
Log likelihood	-3.012993	Hannan-Quinn criter.	0.312144
F-statistic	33.43515	Durbin-Watson stat	0.984937
Prob(F-statistic)	0.000000		

La variable de impacto, junto con la tendencia y el intercepto resultó significativa



Null Hypothesis: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.778555	0.0003
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

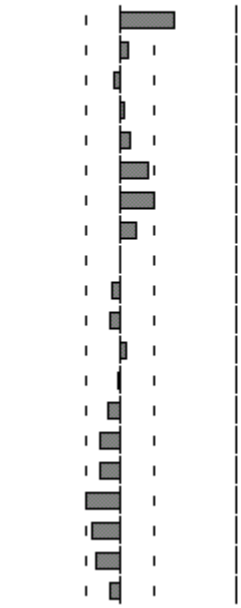
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
)
Method: Least Squares
Date: 11/02/26 Time: 13:18
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INCENTIVO_CONTRABANDO_VI_SIN_TE	-0.502591	0.133011	-3.778555	0.0005

R-squared	0.249257	Mean dependent var	-0.001013
Adjusted R-squared	0.249257	S.D. dependent var	0.262672
S.E. of regression	0.227593	Akaike info criterion	-0.100047
Sum squared resid	2.227348	Schwarz criterion	-0.059497
Log likelihood	3.201025	Hannan-Quinn criter.	-0.085009
Durbin-Watson stat	1.814054		

Date: 11/02/26 Time: 13:18
Sample: 1980 2024
Included observations: 45



	AC	PAC	Q-Stat	Prob
1	0.483	0.483	11.235	0.001
2	0.074	-0.209	11.502	0.003
3	-0.066	-0.010	11.719	0.008
4	0.038	0.131	11.793	0.019
5	0.088	-0.003	12.203	0.032
6	0.256	0.274	15.758	0.015
7	0.302	0.097	20.824	0.004
8	0.141	-0.075	21.960	0.005
9	0.002	0.041	21.960	0.009
10	-0.081	-0.116	22.358	0.013
11	-0.082	-0.052	22.778	0.019
12	0.047	0.104	22.921	0.028
13	-0.015	-0.281	22.935	0.042
14	-0.102	-0.044	23.649	0.050
15	-0.181	-0.103	25.955	0.038
16	-0.185	-0.153	28.448	0.028
17	-0.312	-0.202	35.784	0.005
18	-0.258	-0.075	40.984	0.002
19	-0.219	-0.208	44.876	0.001
20	-0.092	0.147	45.592	0.001

Dependent Variable: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:19
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 3 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.497409	0.133011	3.739602	0.0005

R-squared	0.245059	Mean dependent var	0.005592
Adjusted R-squared	0.245059	S.D. dependent var	0.261941
S.E. of regression	0.227593	Akaike info criterion	-0.100047
Sum squared resid	2.227348	Schwarz criterion	-0.059497
Log likelihood	3.201025	Hannan-Quinn criter.	-0.085009
Durbin-Watson stat	1.814054		

Inverted AR Roots	.50

Date: 11/02/26 Time: 13:20
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.082	0.082	0.3154	
		2	-0.173	-0.181	1.7550	0.185
		3	-0.136	-0.109	2.6723	0.263
		4	0.036	0.027	2.7362	0.434
		5	-0.041	-0.092	2.8229	0.588
		6	0.155	0.171	4.1060	0.534
		7	0.253	0.231	7.6212	0.267
		8	-0.012	-0.015	7.6297	0.366
		9	-0.049	0.090	7.7662	0.457
		10	-0.052	-0.021	7.9283	0.541
		11	-0.080	-0.103	8.3241	0.597
		12	0.143	0.181	9.6105	0.566
		13	-0.023	-0.197	9.6463	0.647
		14	-0.061	-0.080	9.8933	0.703
		15	-0.080	-0.043	10.337	0.737
		16	0.038	-0.067	10.439	0.791
		17	-0.236	-0.242	14.605	0.554
		18	-0.056	-0.032	14.849	0.606
		19	-0.088	-0.295	15.480	0.629
		20	-0.042	-0.050	15.629	0.682

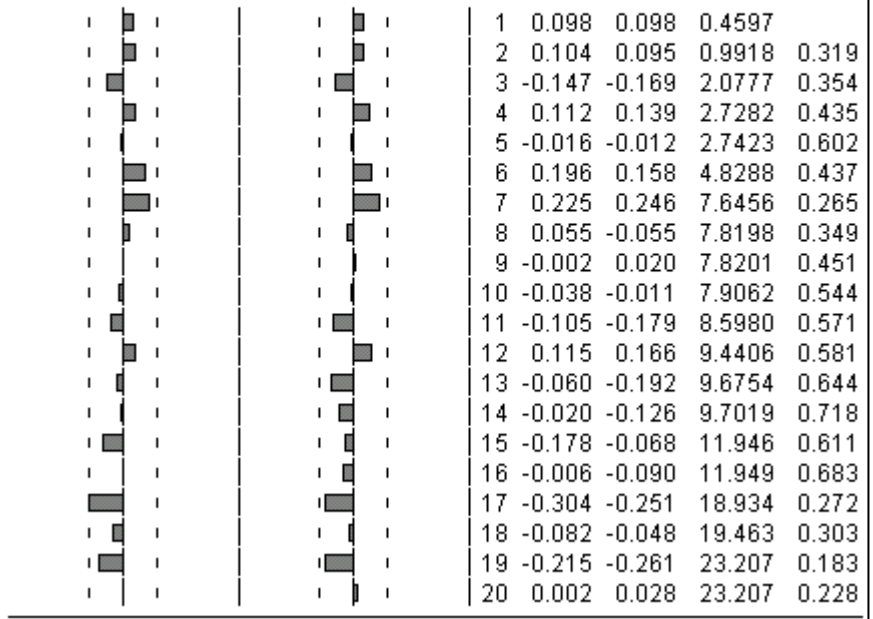
Dependent Variable: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/02/26 Time: 13:20
Sample: 1980 2024
Included observations: 45
Convergence achieved after 42 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.478994	0.150264	3.187685	0.0027
SIGMASQ	0.050568	0.008774	5.763403	0.0000

R-squared	0.244574	Mean dependent var	-5.43E-17
Adjusted R-squared	0.227006	S.D. dependent var	0.261650
S.E. of regression	0.230043	Akaike info criterion	-0.051882
Sum squared resid	2.275551	Schwarz criterion	0.028414
Log likelihood	3.167355	Hannan-Quinn criter.	-0.021949
Durbin-Watson stat	1.771654		

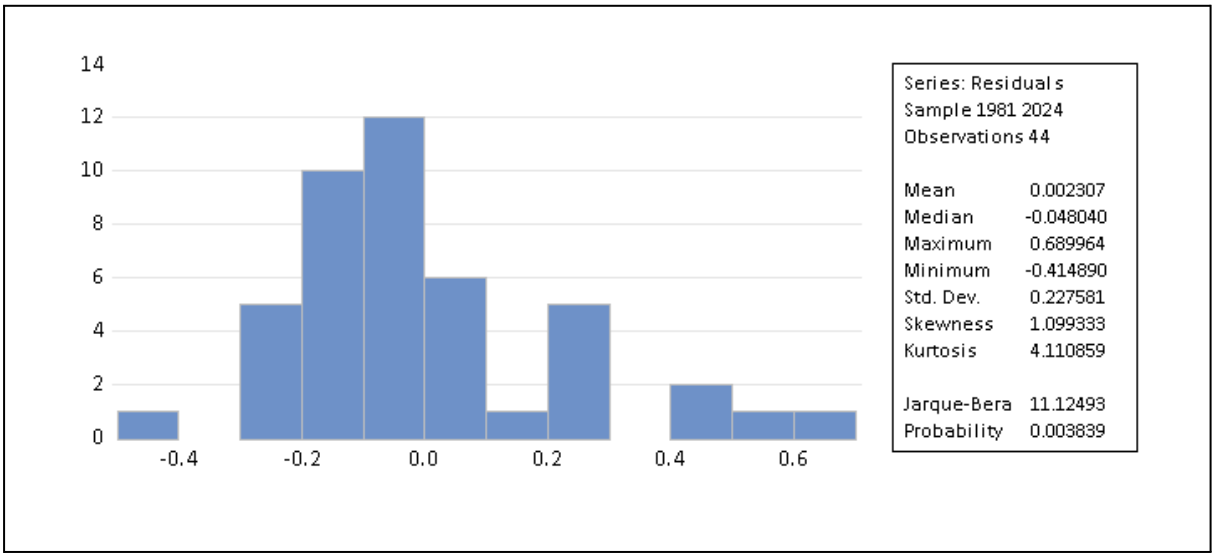
Inverted MA Roots	-.48

Date: 11/02/26 Time: 13:21
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term



	Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
1			0.098	0.098	0.4597	
2			0.104	0.095	0.9918	0.319
3			-0.147	-0.169	2.0777	0.354
4			0.112	0.139	2.7282	0.435
5			-0.016	-0.012	2.7423	0.602
6			0.196	0.158	4.8288	0.437
7			0.225	0.246	7.6456	0.265
8			0.055	-0.055	7.8198	0.349
9			-0.002	0.020	7.8201	0.451
10			-0.038	-0.011	7.9062	0.544
11			-0.105	-0.179	8.5980	0.571
12			0.115	0.166	9.4406	0.581
13			-0.060	-0.192	9.6754	0.644
14			-0.020	-0.126	9.7019	0.718
15			-0.178	-0.068	11.946	0.611
16			-0.006	-0.090	11.949	0.683
17			-0.304	-0.251	18.934	0.272
18			-0.082	-0.048	19.463	0.303
19			-0.215	-0.261	23.207	0.183
20			0.002	0.028	23.207	0.228

Comparo ambos modelos según el criterio de Akaike, en donde gana el modelo con AR(1) (-0.1<-0.05)



Comparado con el modelo sin variable de intervención se ve que mejora mucho el criterio de información y disminuye la varianza, además a pesar de que el test de normalidad sigue siendo rechazado tiene mucha menor asimetría y curtosis que el modelo anterior.